



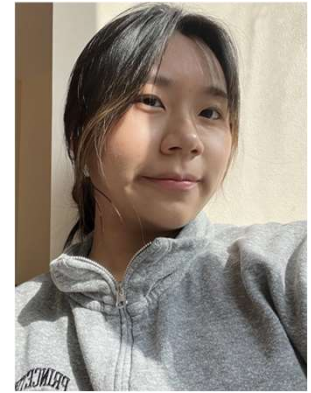
Critical Design Review

Ultra-thin Carbon Foil Analyzer

**AST 250/251
Spring 2024
Space Physics at Princeton**



AST 250/251 Team (2023-2024)





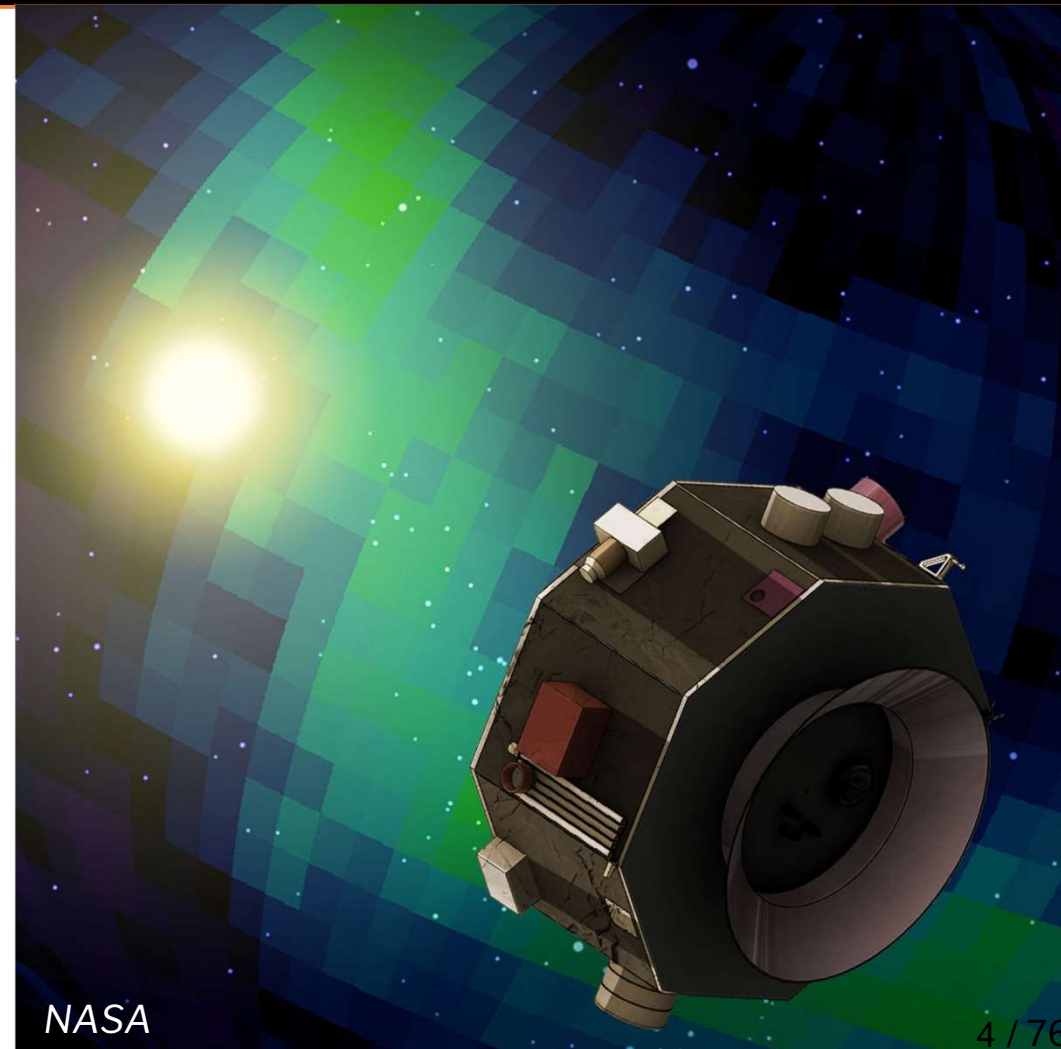
Critical Design Review

Background



Interstellar Mapping and Acceleration Probe (IMAP)

- Investigating the heliosphere:
 - Composition of the Local Interstellar Medium (LISM)
 - Interactions between Solar Wind and the LISM
 - Interactions between the solar magnetic fields and the LISM
 - Particle injection and acceleration

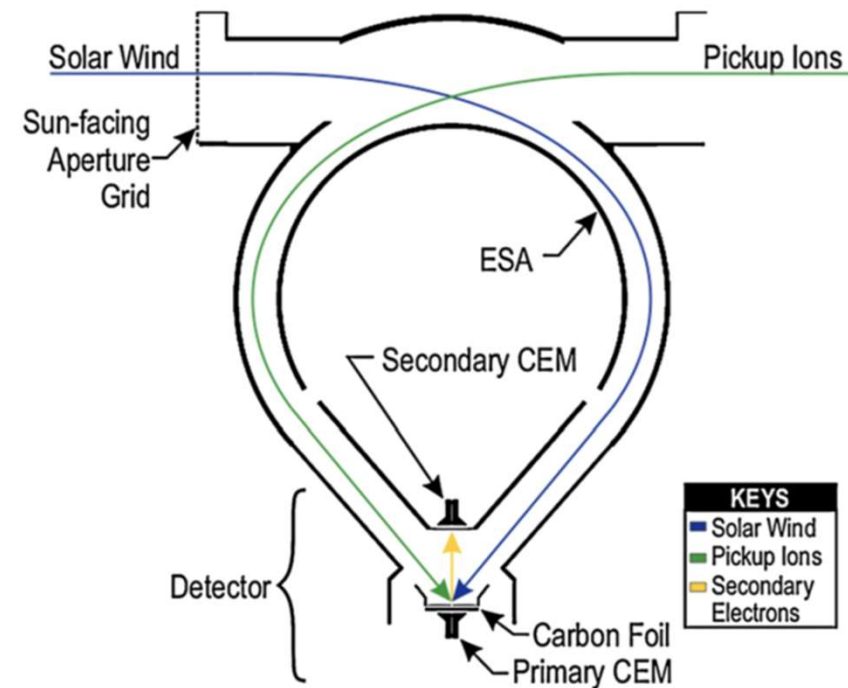




Solar Wind and Pickup Ions (SWAPI)

- Measures elements of solar wind
- Interstellar pickup ions (PUIs)
- Space weather monitoring
 - Solar wind changes over time
- High time resolution measurements of helium PUIs
- Contains carbon foil for coincidence electron

testing



Princeton/IMAP

Motive for Testing

- Installation process for foil creates potential for defects/rips
 - Density of $1 \mu\text{g}/\text{sq.cm}$ (± 0.2) and thinness of foil
- Need to characterize carbon foil thickness variation and scattering

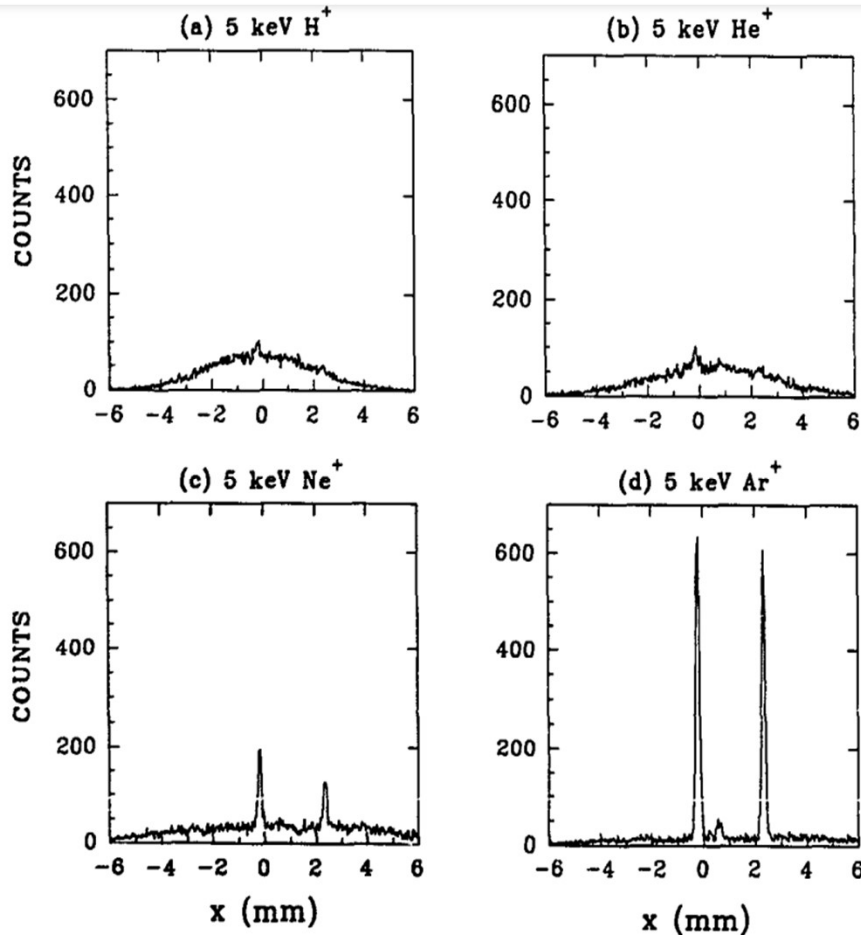


Motive for Testing

- Crud layer
 - Composed of water vapor and other “crud”
 - Local thickness measurements of foil would measure foil thickness with crud
 - Testing could occur on how thickness of crud varies properties of foil

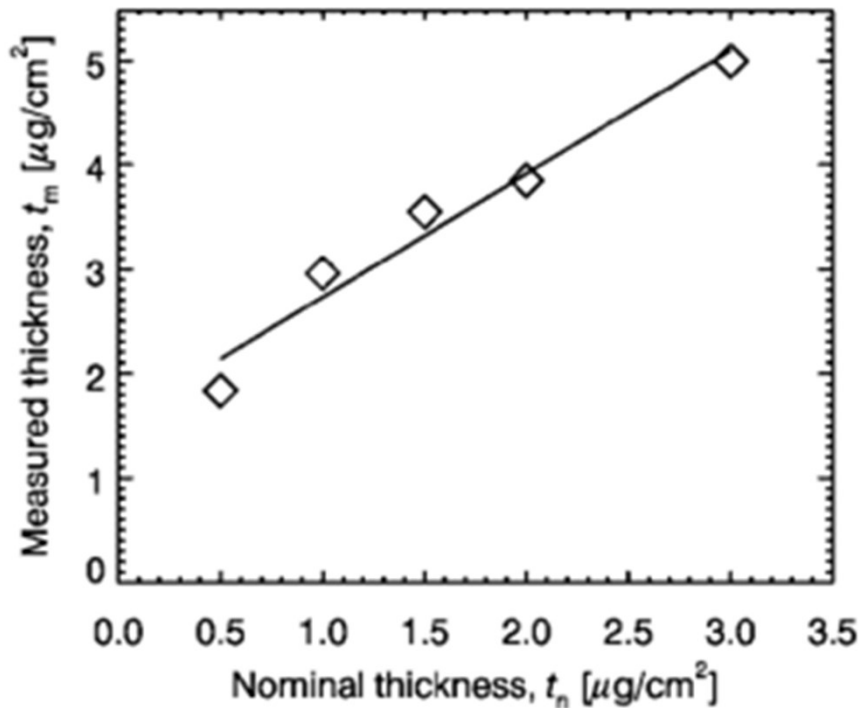


Previous Findings: Pinholes and Thin Defects



- Shoot ions through the bulk foil to obtain a two component distribution:
 - Broader scattering angle distribution, which is interesting on its own (see next slide)
 - Deviations from that distribution reflecting surface features such as pinholes and extremely thin defects
- Funsten et al. 1991 found that identifying pinholes is easier for heavier ions
- These 1D images are slices from a 2D image

Previous Findings: Bulk Thickness



- Measure scattering angle half-width and use following relationship to determine foil constant:

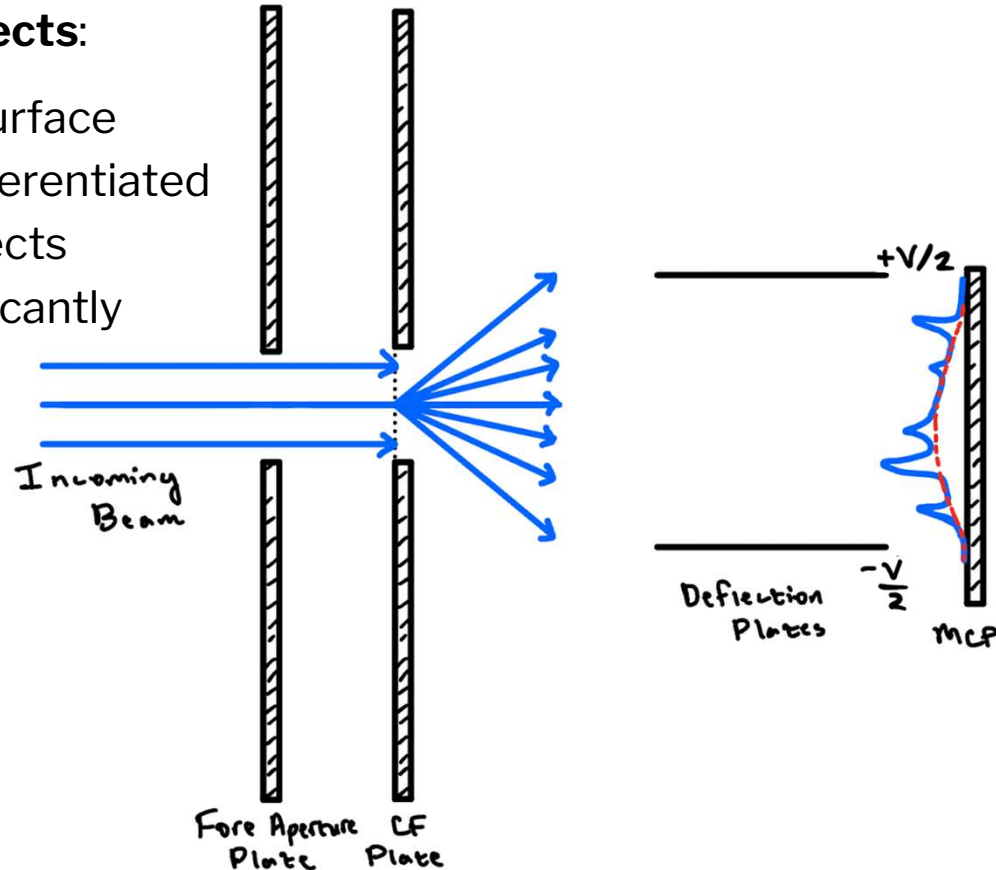
$$E = \frac{k_F}{\psi_{1/2}}$$
- Can then obtain the foil thickness using Meyer's theory

$$t(\mu\text{g cm}^{-2}) = \frac{\bar{E}(\text{keV})\psi_{1/2}(\text{deg})}{12Z^{3/4}}$$

Experiment 1

(1) Pinholes and extremely thin defects:

- Past studies* have produced surface maps of features but never differentiated between pinholes and thin defects
- Perhaps thin defects are significantly more effective than pinholes?



*e.g. Funsten et al 1992

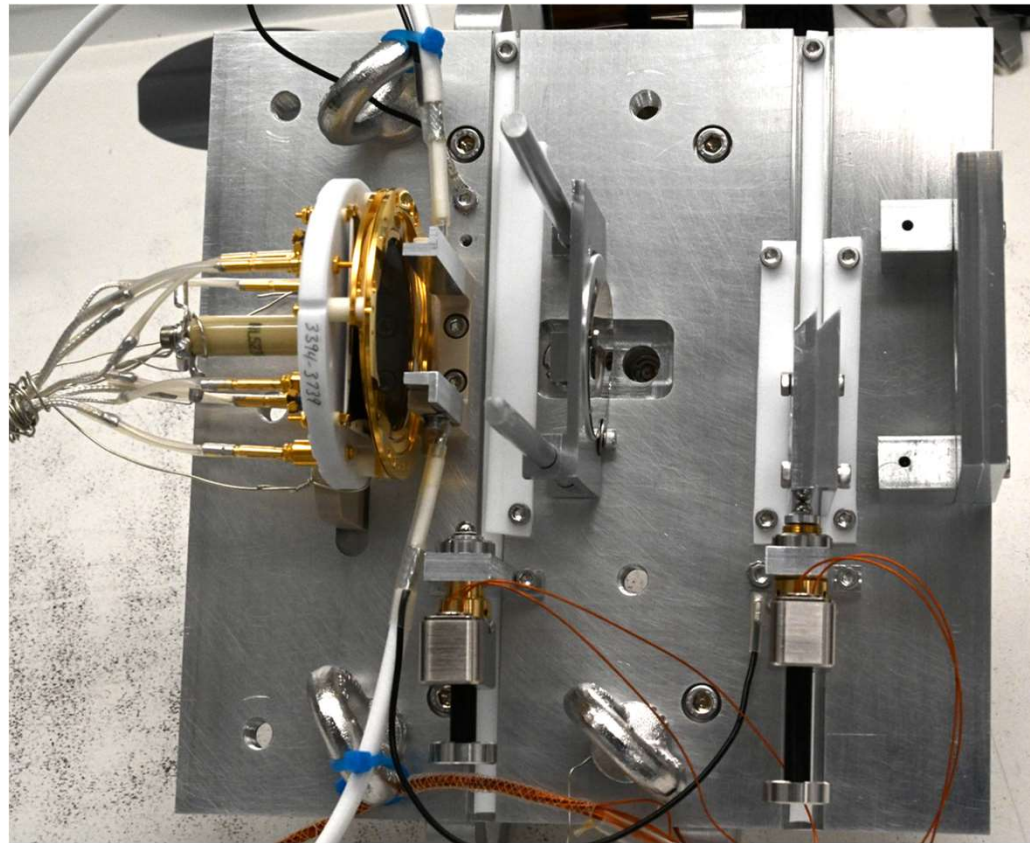


Critical Design Review

Device Overview

Carbon Foil Analyzing Apparatus

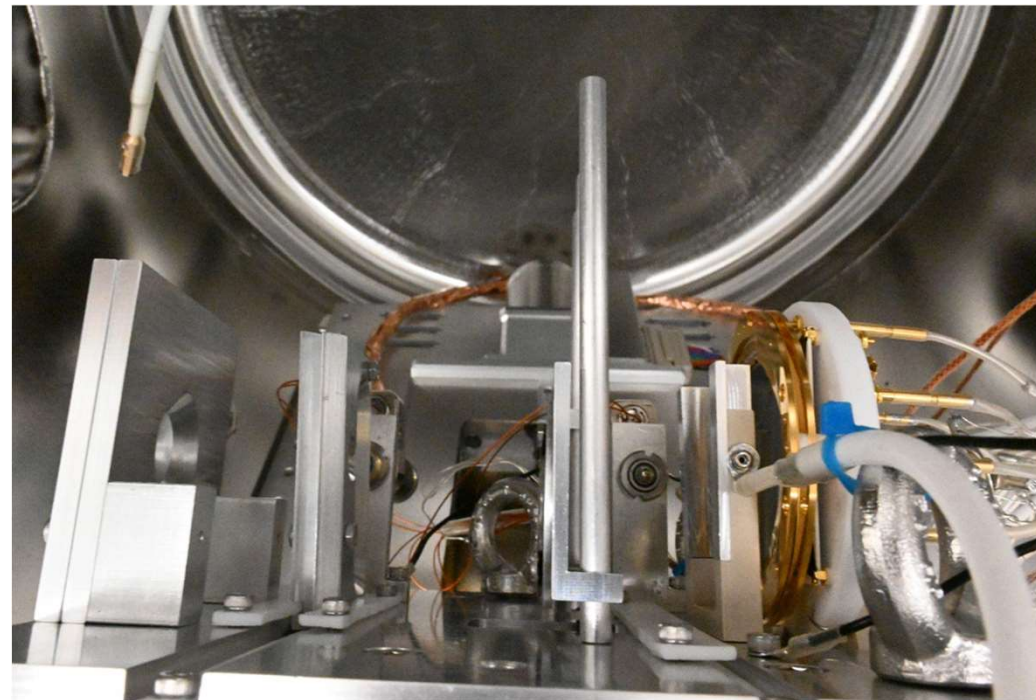
- Goal: to pass an ion beam through a carbon foil and detect the ion produced
- Beam passes:
 - Through a fixed aperture
 - Through a moveable plate with a pinhole and knife edge
 - Through a moveable carbon foil
 - Optionally, through a moveable plate with a slit and pinhole
 - Between two parallel plates at opposite high voltage
 - Onto the MCP face



Carbon Foil Analyzing Apparatus

Allows us to:

- Analyze the full half-foil or a specific point
- Analyze all ions/neutrals on MCP or deflect ions away to analyze only neutrals
- Separate charge state populations on MCP face using slit on the aft plate
- Block all but a specific scattering angle on MCP using pinhole on the aft plate



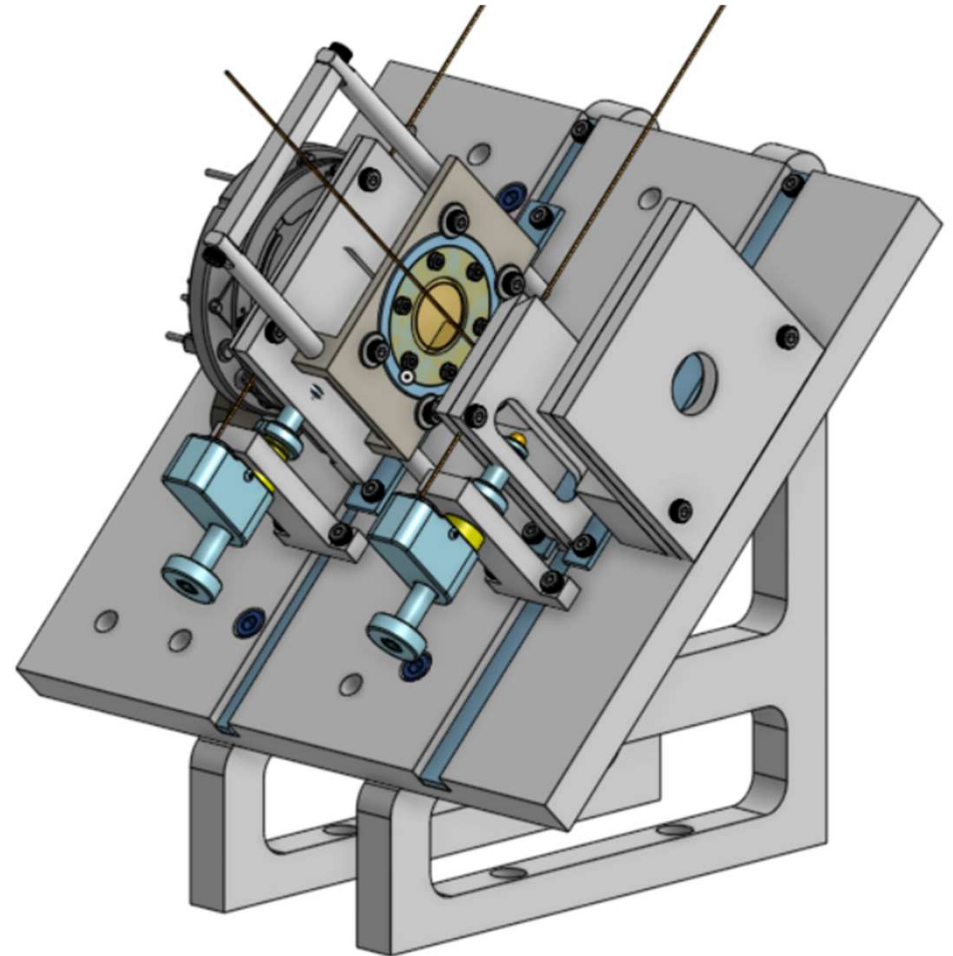


Critical Design Review

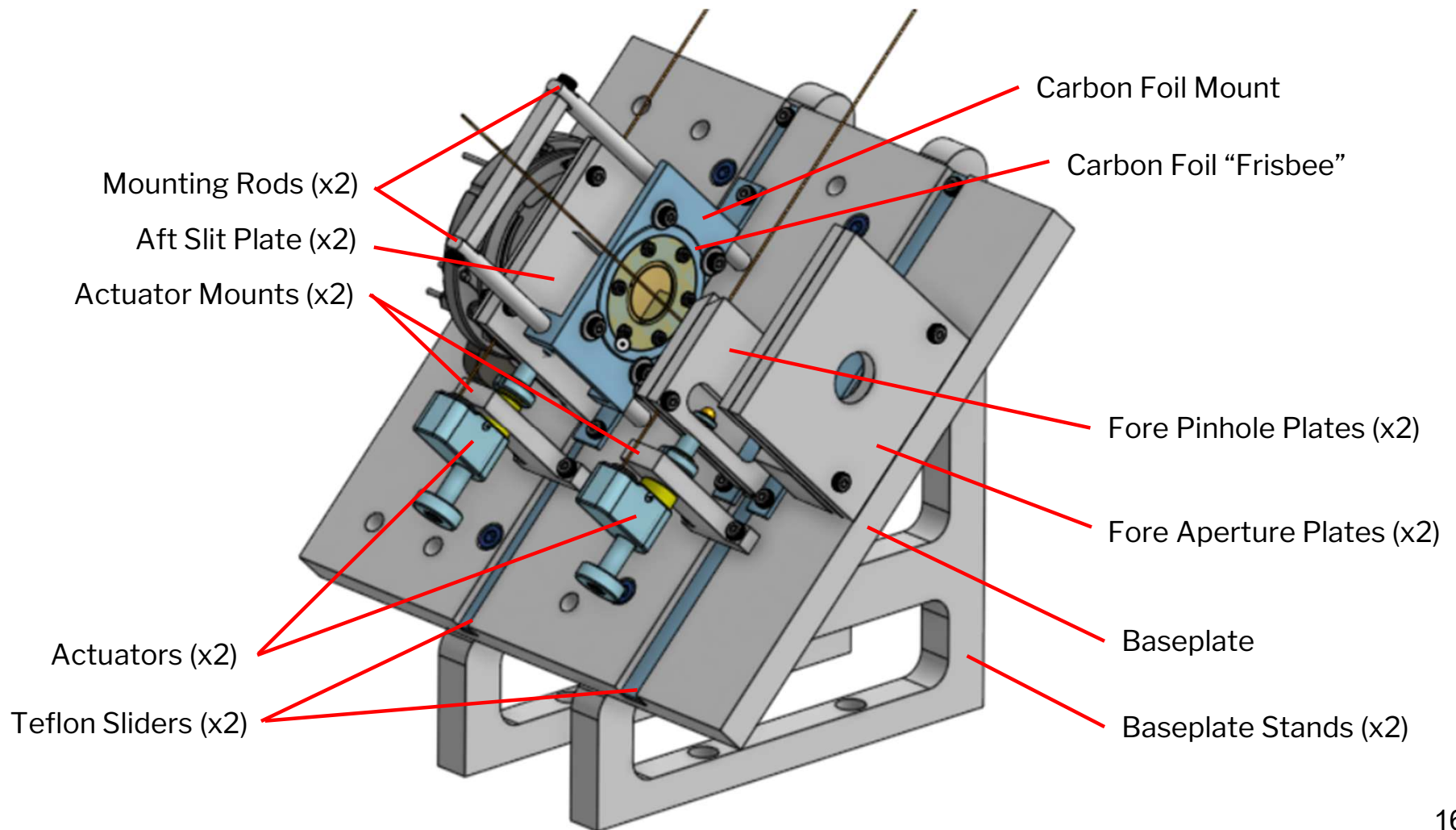
Changes Since PDR

Full Assembly

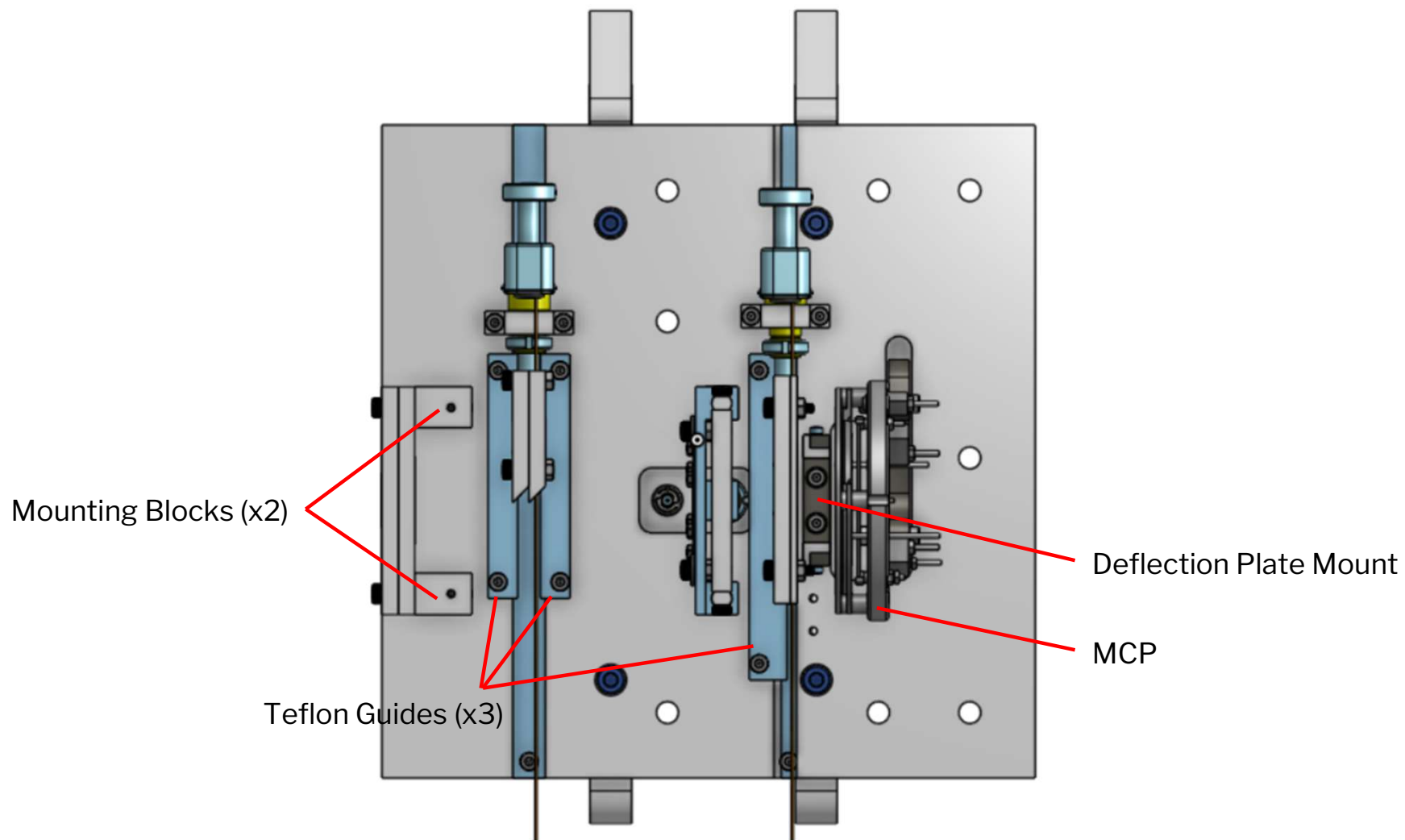
- Added stands to tilt apparatus to 45 degree angle
- Actuators cannot attach to plates; gravity provides restoring force
- Ensures contact with plates and actuator at all times



Part Locations



Part Locations



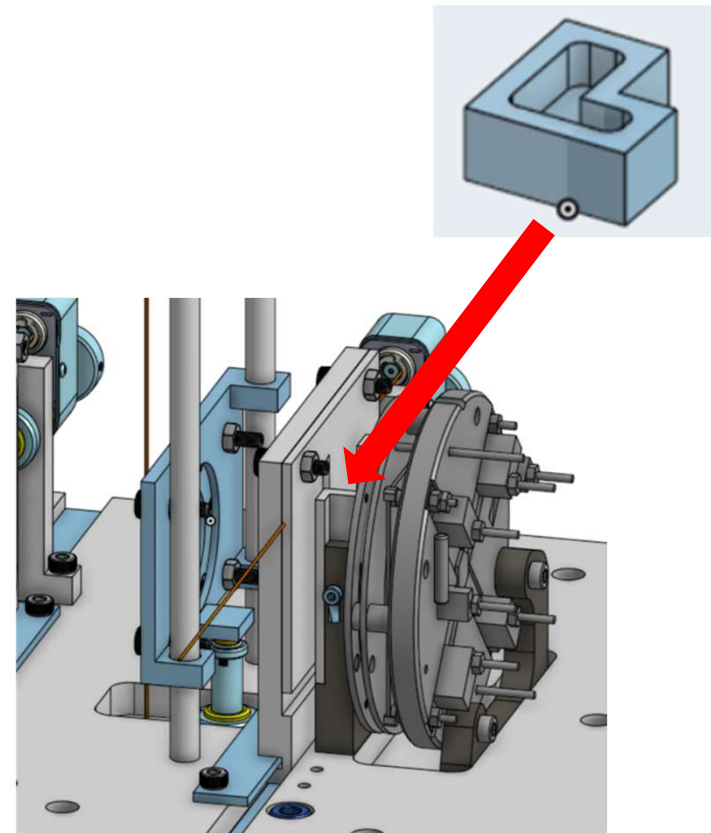
Baseplate

- Grooves were too narrow for plates to slide in them
- Enlarged with sandpaper to ensure smooth sliding



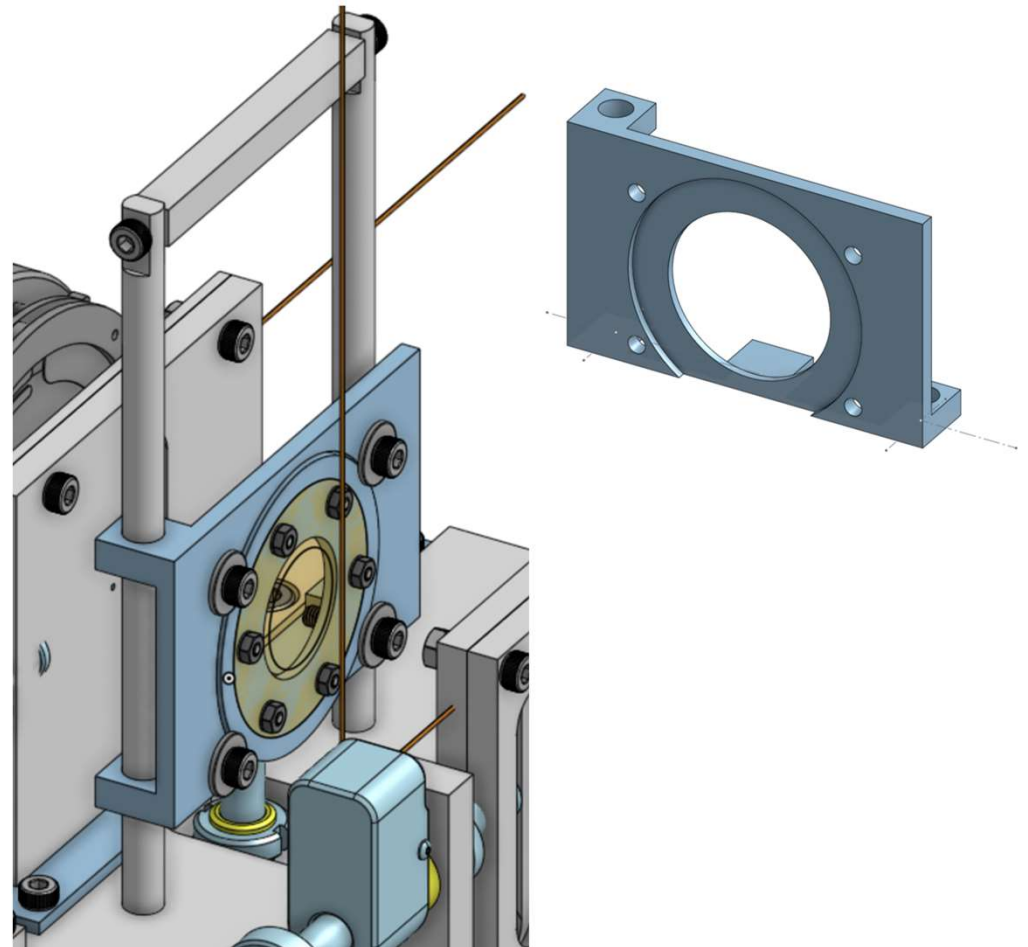
Teflon Endcaps

- Tolerances of baseplate slits and aft slit plate allow aft slit plate to touch deflector plates
- Two teflon endcaps designed to prevent arcing
- Order in progress if future testing with apparatus occurs



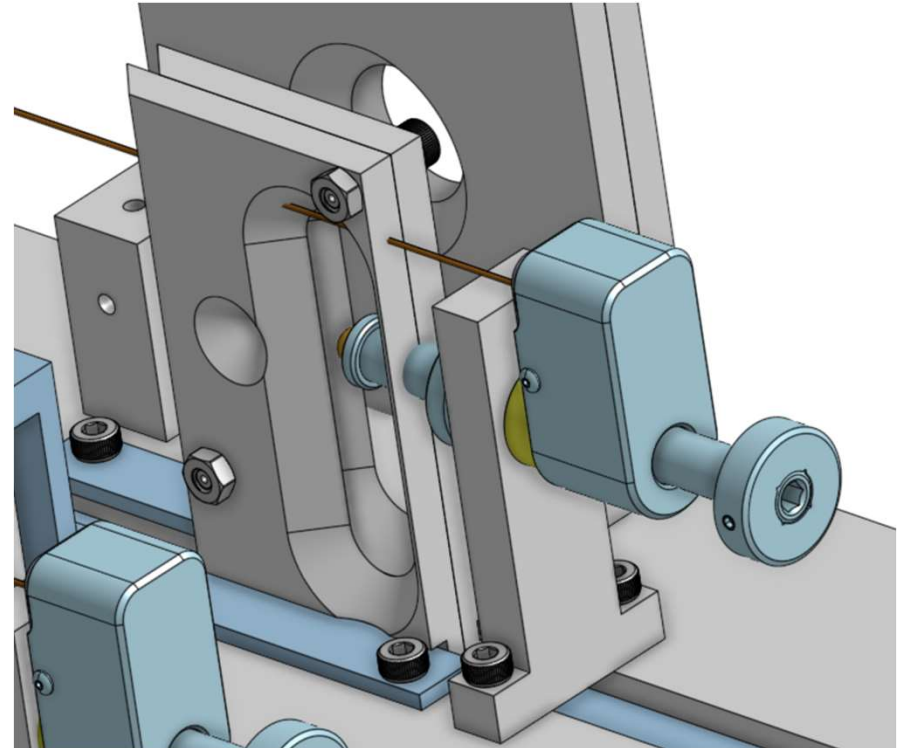
New CF Mount Design

- Circular recess allows for rotation of foil in case of partial floating
- Carbon foil mount secured with 4 washers
- Carbon foil assembly mounted on two smooth rods for vertical motion
- Actuator pushes from underneath



Knife Edges

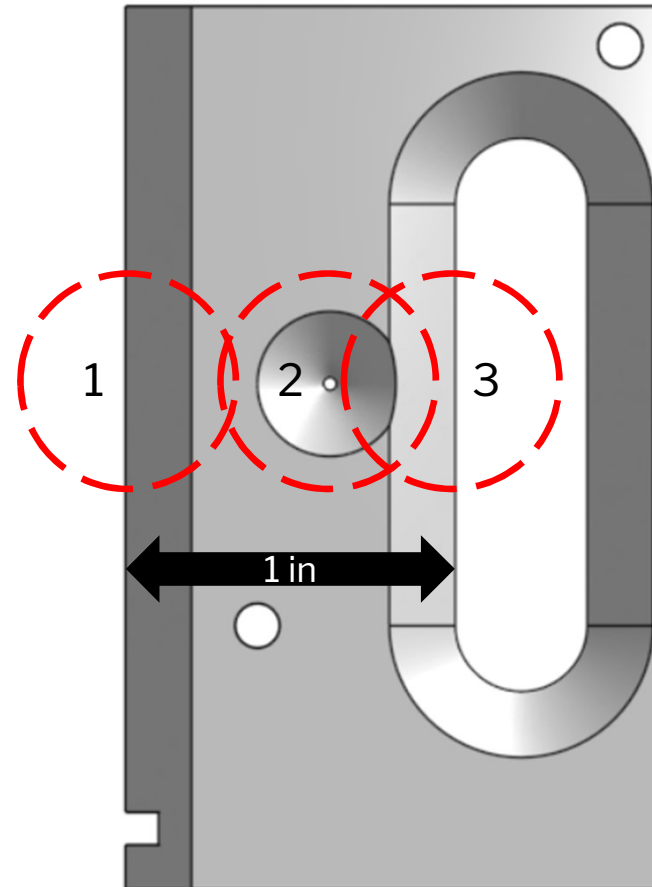
- Implemented knife edges to collimate beam through pinhole and slit
- Secured in parallel with 2 M3 screws and hex nuts
- Fore aperture knife edge does not collimate beam much



Fore Pinhole Plate

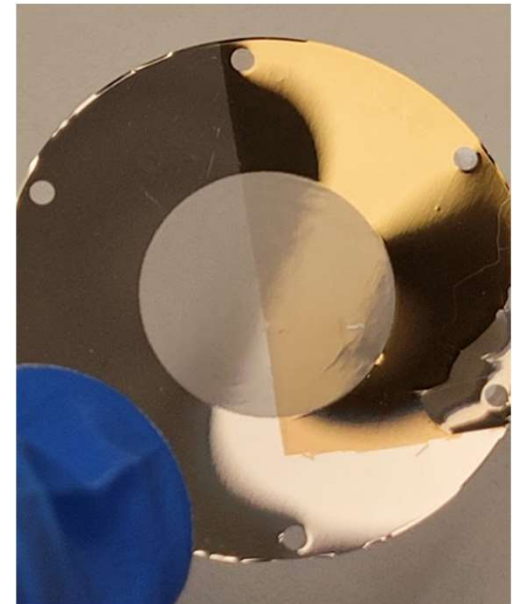
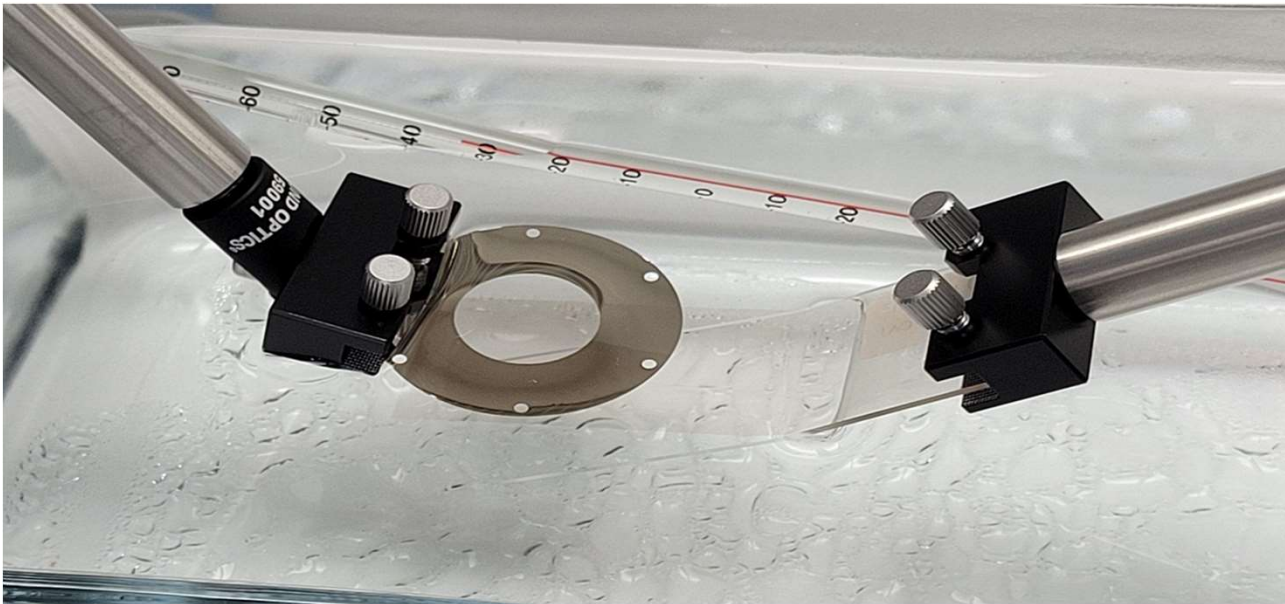
3 possible positions allow for different areas of foil grid to be exposed to beamline

1. Left side of beam to test covered/uncovered half of foil grid
2. Pinhole to test scattering across foil
3. Right side of beam to test uncovered/covered half of foil grid



Half-Floated Carbon Foil Grid

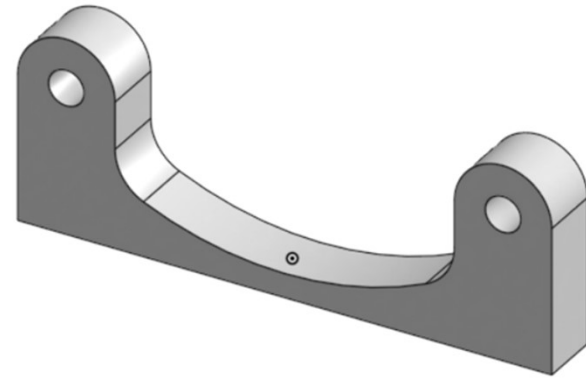
Only half the grid is covered to allow for a control measurement



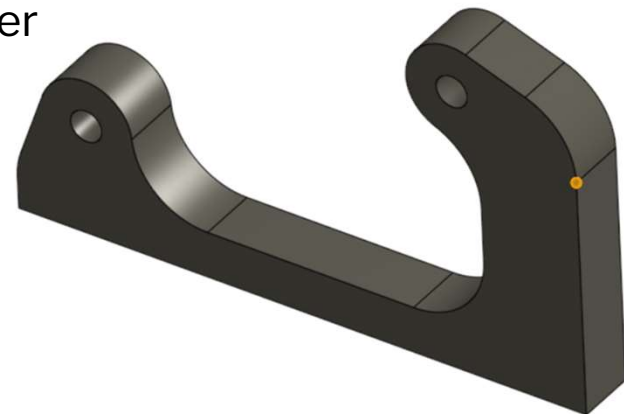
New MCP Mount

- Positions of holes rotated to orient MCP in “upright” position
- CNC machined from PEEK

Before

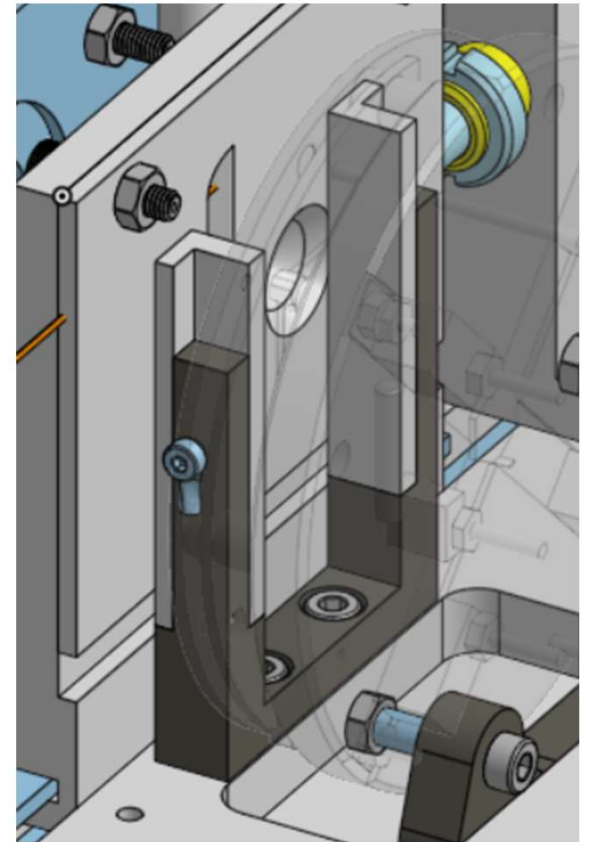
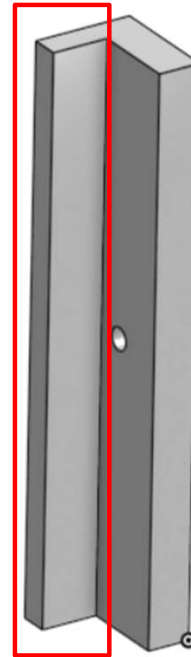


After



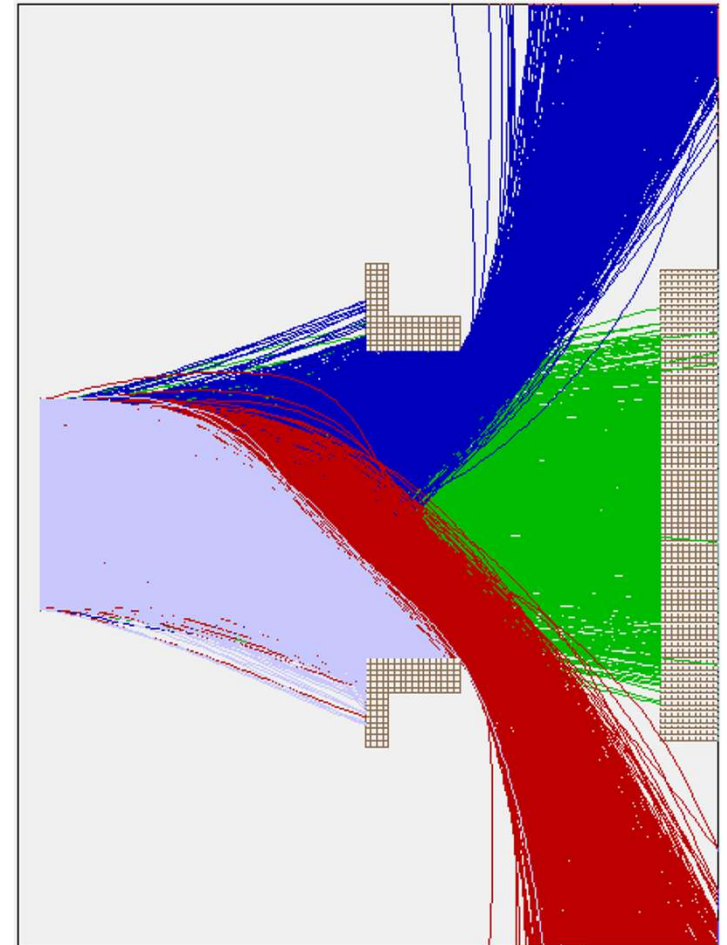
New Deflection Plates

- Extra protrusion added to face beam line
- Covers and prevents charge buildup on PEEK deflection plate mount
- Threaded hole for M2 screw and ring connector for electrical connection

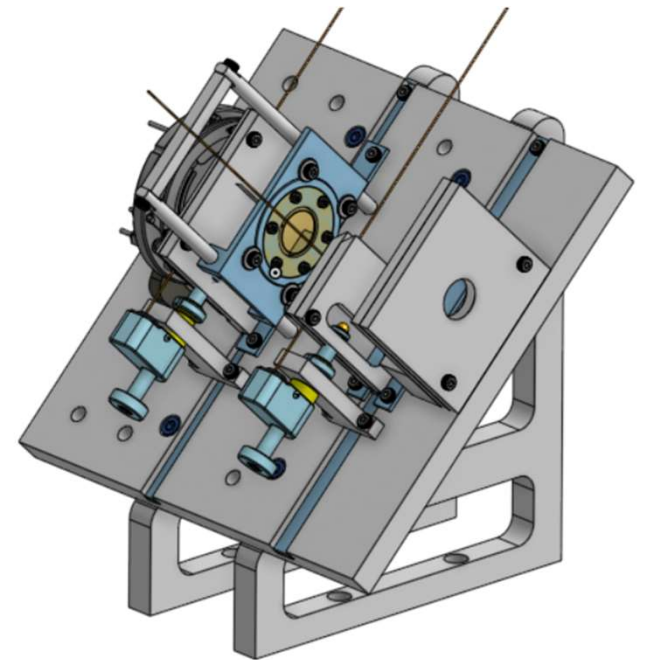
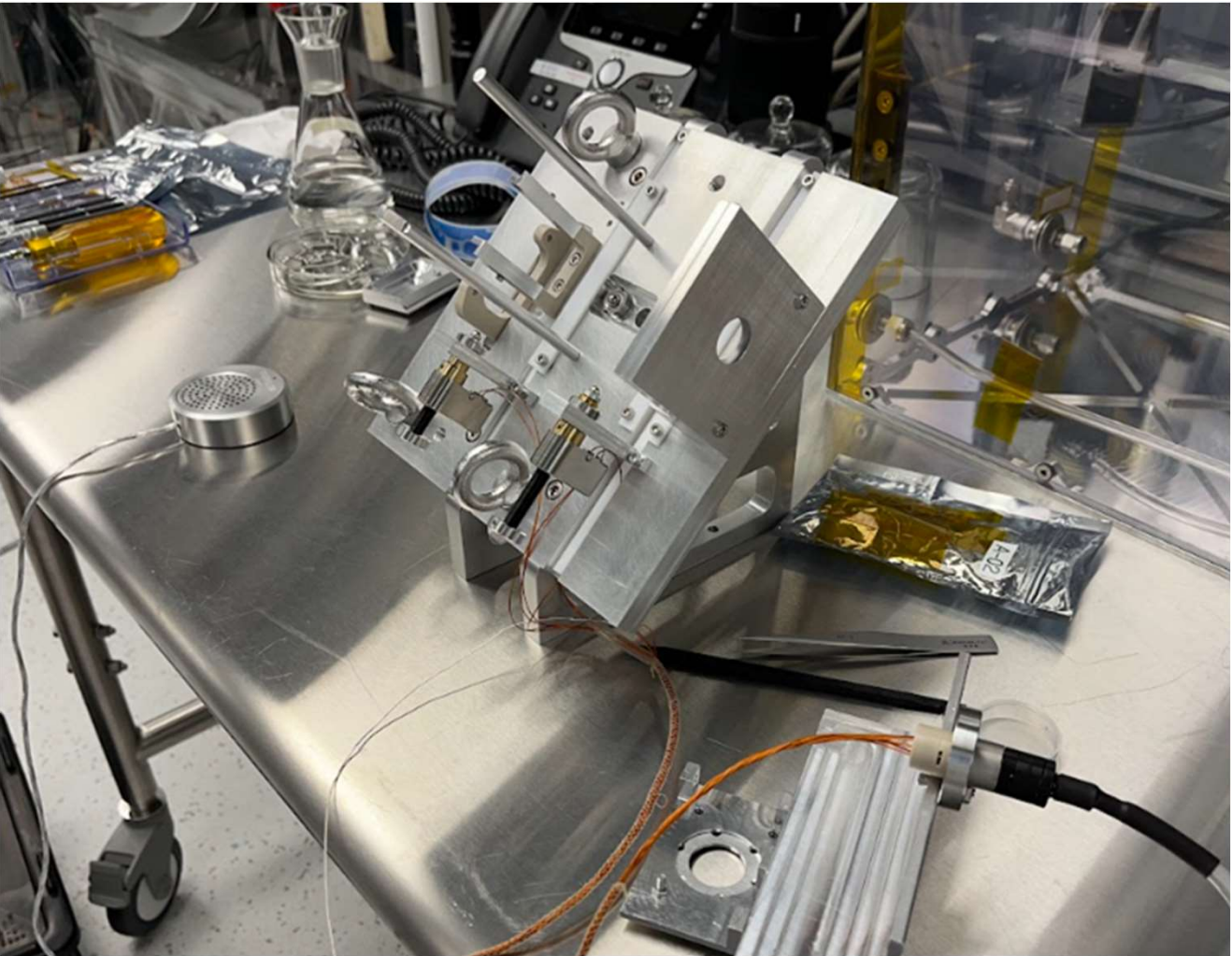


New Deflection Plates

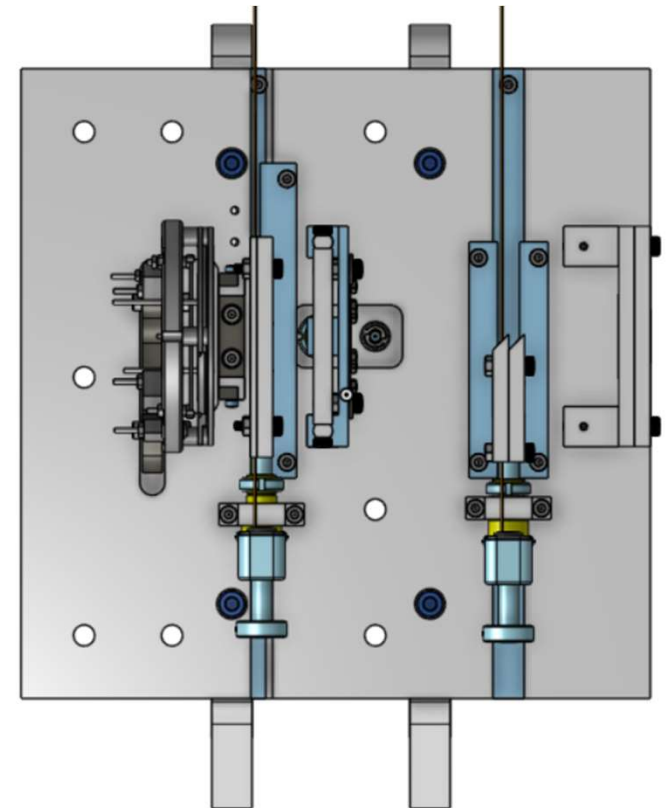
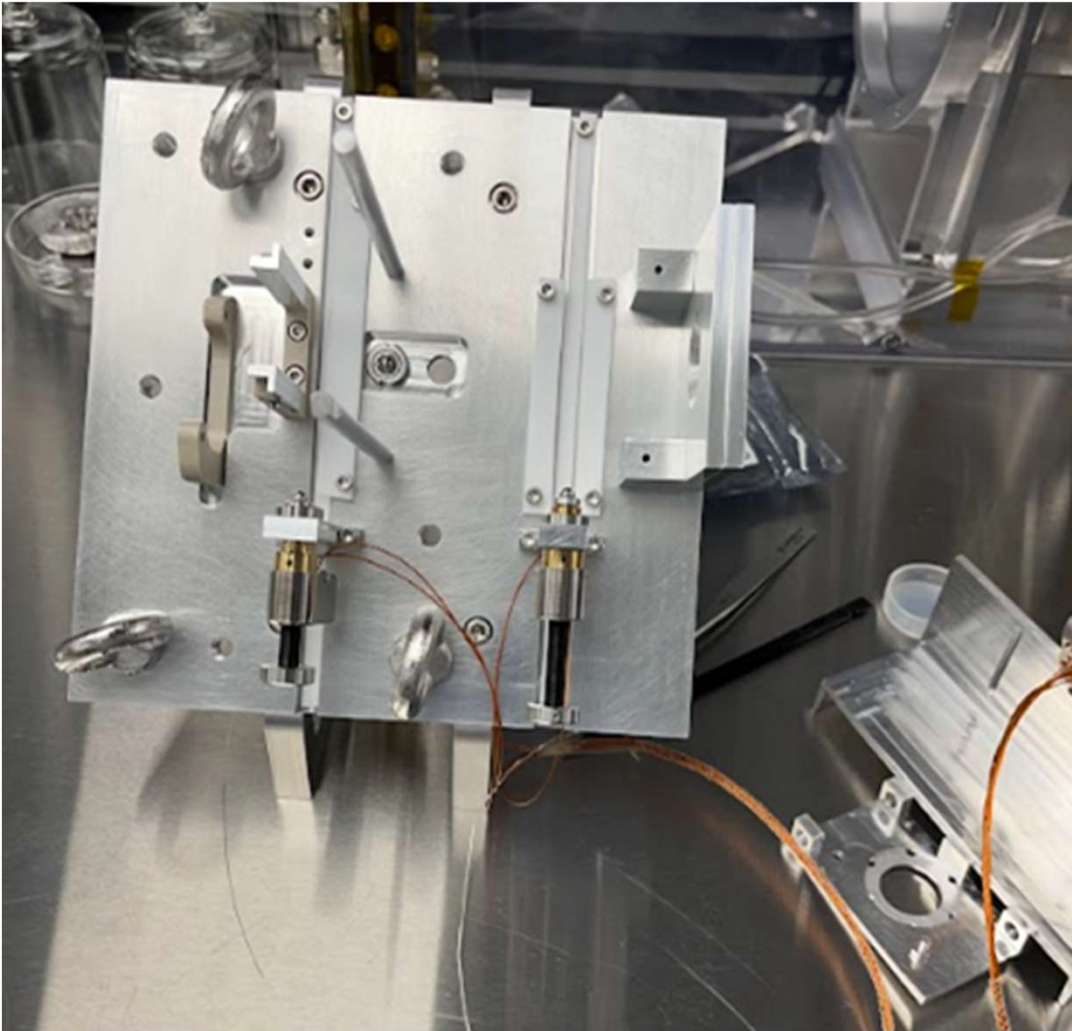
- Tested new design in SIMION
 - 1.5 KeV Argon
- Constraints were placed on deflection plate voltage based on separation from other parts



Pictures



Pictures





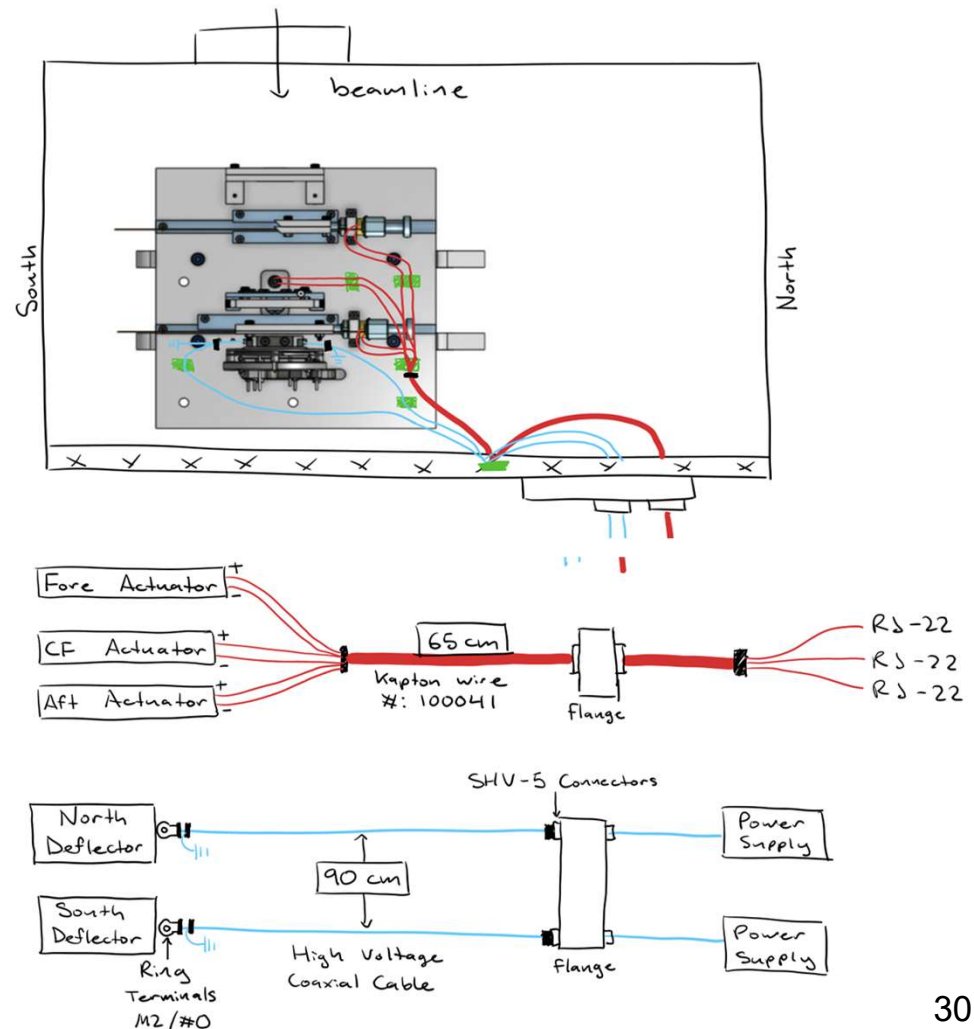
Critical Design Review

Electronics

Electronics

There are 3 electronics systems in the apparatus:

1. MCP
 - a. Wiring to be done by lab
 - b. Special considerations: The back wires are unshielded HV lines
2. Deflector Plates
 - a. Grounding: through base plate to chamber, kept far from rear of MCP
 - b. HV safety: Insulators and caps
 - c. Signal path:
3. Actuators:
 - a. Signal path: 6x30 gauge → 9 pin → RJ-22
 - b. Grounding was tied under a ring bolt



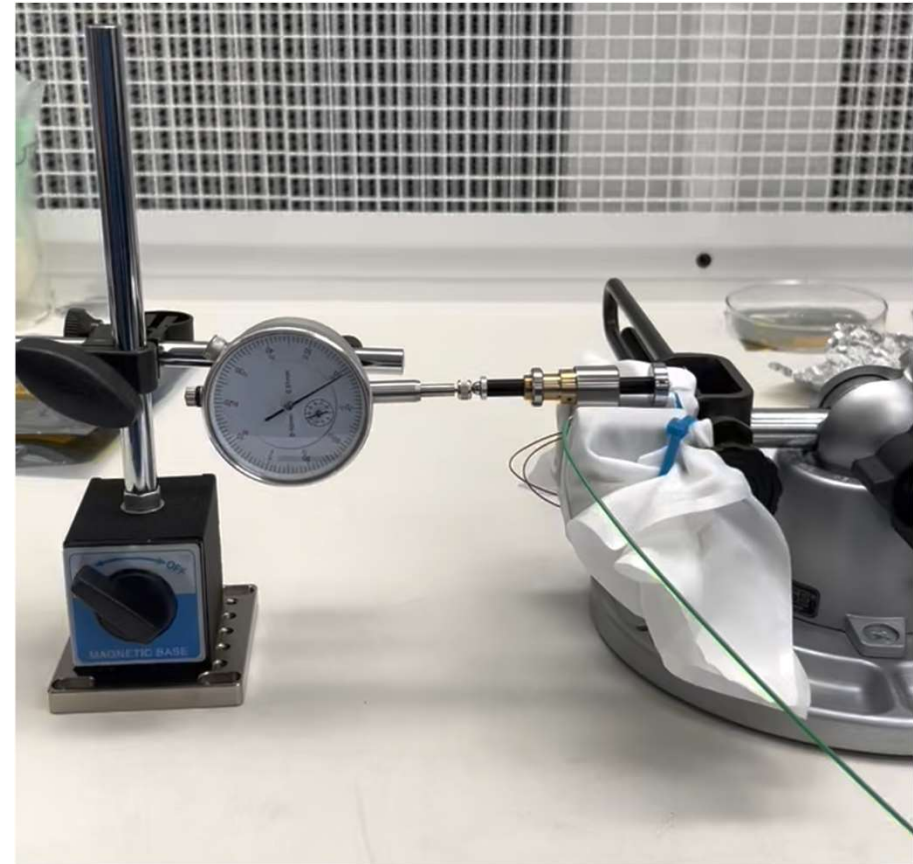
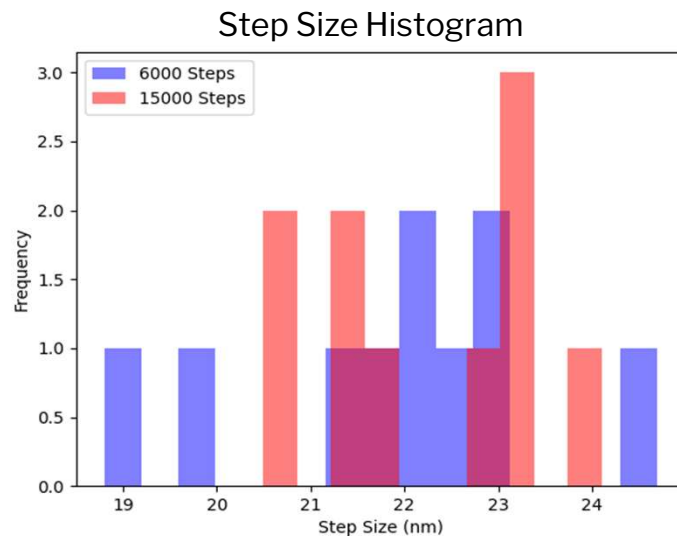


Critical Design Review

Calibration

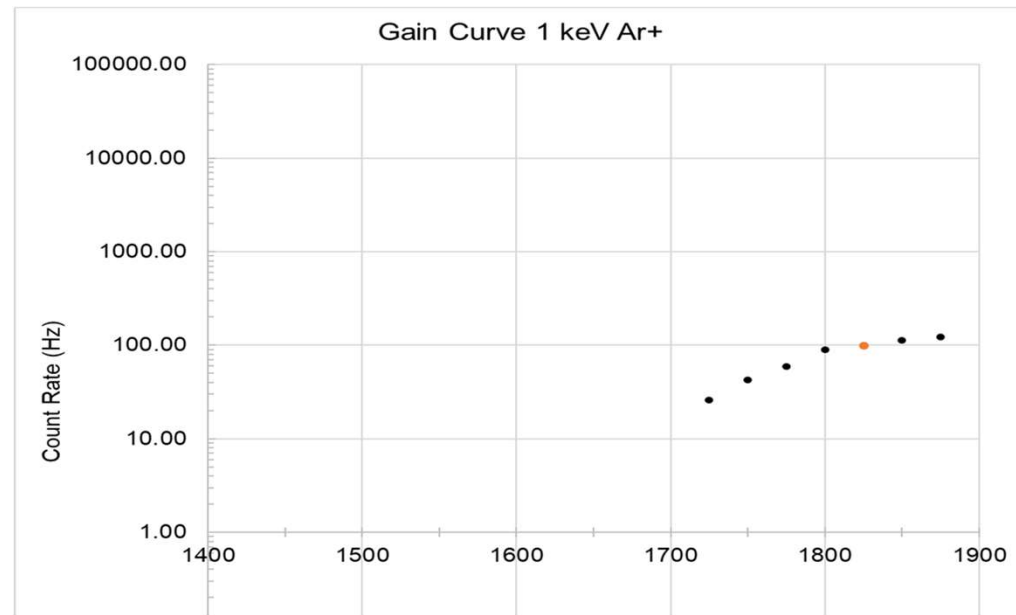
Actuators

- Our actuators are open-loop; don't know absolute position
 - Used force gauge to estimate mean step size
 - Result: 22.07 ± 0.63 nm



MCP - Gain Curves

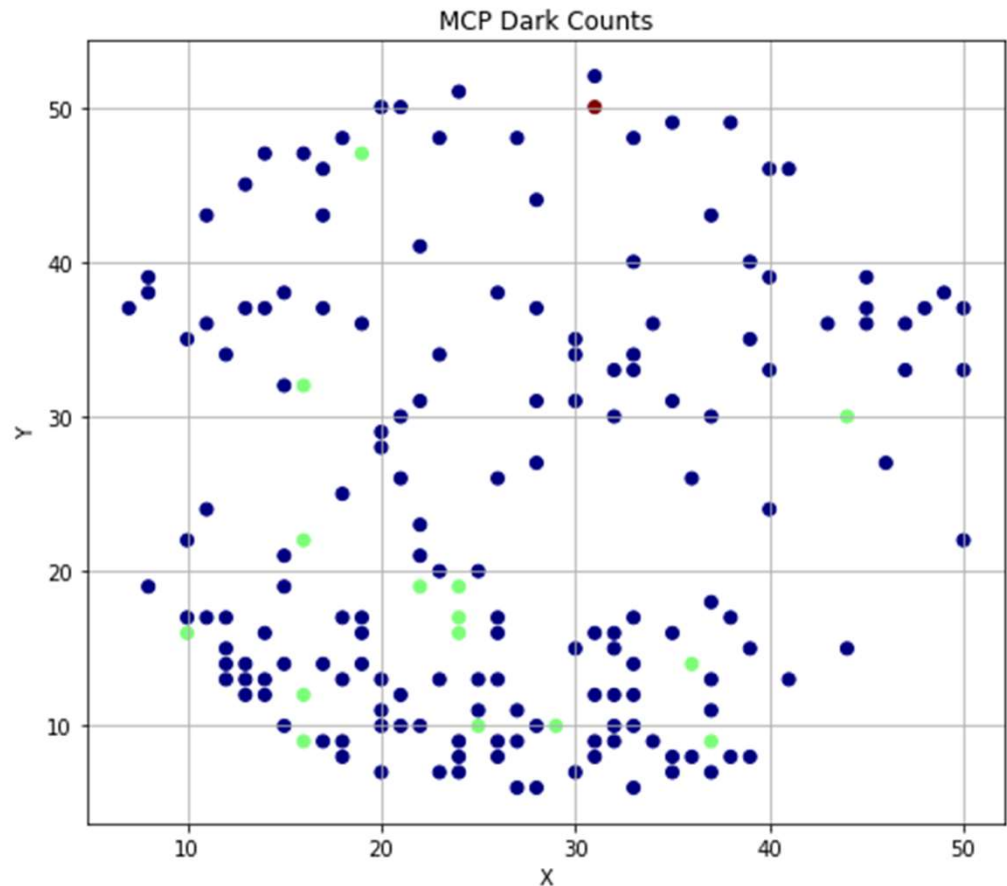
- We planned to obtain several gain curves for different beams that we needed for our experiment
- We decided to use a heavy particle (Ar^+) and a light particle (H^+) so we can have different FWHMs in our kappa distributions; Additionally, we also used He
- And two energies, 1 keV and 1.5 keV, which were determined theoretically at the beginning of the semester
- Ideal voltage for Ar^+ : 1825V



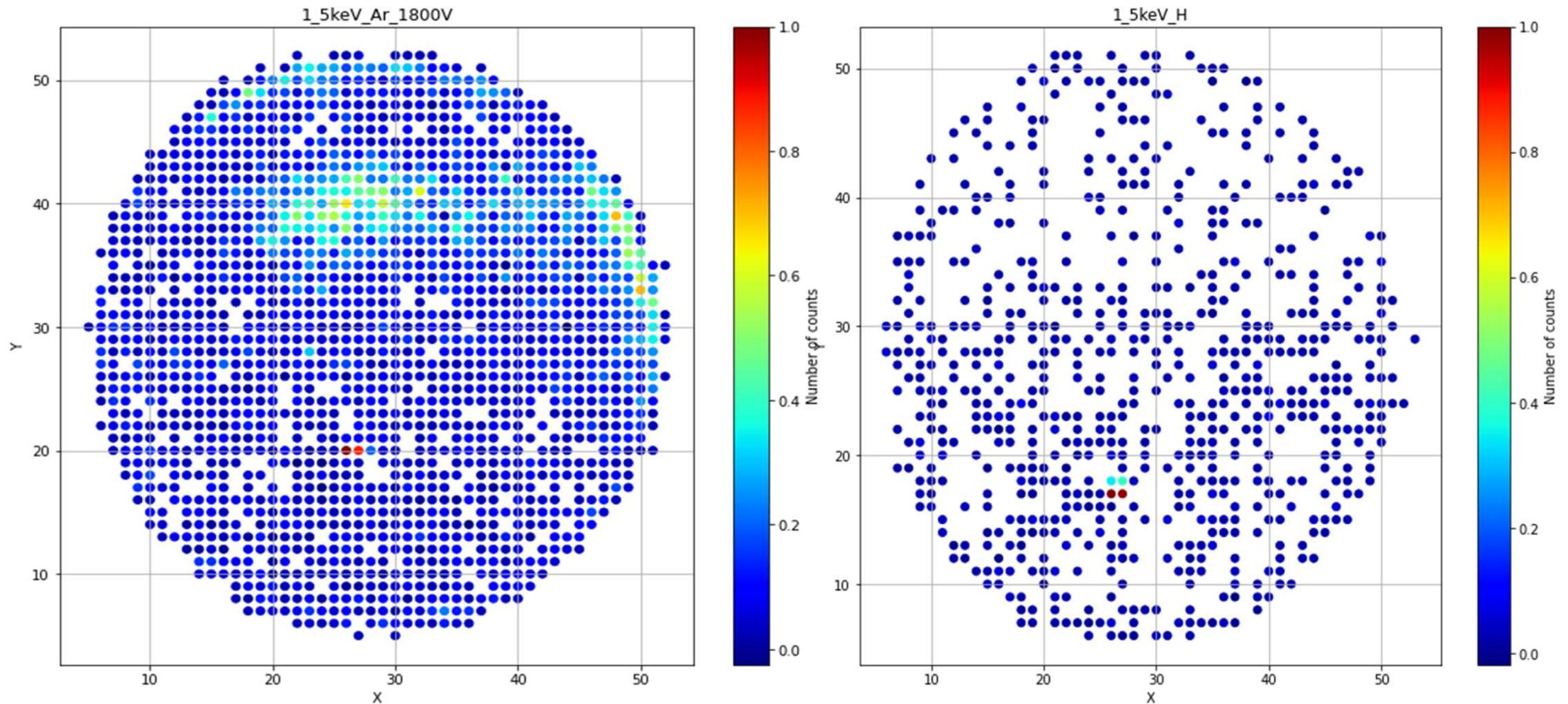


MCP Dark Counts + Beam Imaging

- We also obtained the dark count flat field of the MCP, which we have subtracted from our data to obtain higher accuracy
- We also intended to use the code to plot images of the beam and compare them amongst each other to see if the MCP was picking up how we expected our instrument to behave



Beam Images – Strange Feature



Voltage on Plates = 1000 V (500+500)



Critical Design Review

LabView



Purpose of LabVIEW

- Used to control:
 - Actuators (Fore, Carbon Foil and Aft Actuator)
 - Linear stage (in Science Camber)
 - Voltage Plates
- Actuator LabVIEW
 - Set limits for absolute move
 - Set home position

- Procedure:

https://docs.google.com/document/d/1zRc7Go9DCfkqQmxyp0E_vMy80VJWbdCMbjdEmjJFCv8/edit?usp=sharing



Actuator LabVIEW

Fore Plate Actuator

☐ Fore in Motion?

Selected Controller: 1
8742 P2004 (Address 1)

Selected Motor: Fore
1

Move Relative: Fore

Go

Current Position: Fore

Stop Motion

Status:

Fore Max
450000

Fore Min
0

Fore Plate Position

0 2000 4000 6000

Target Move Fore
0

Carbon Foil Actuator

☐ Foil in Motion?

Selected Controller: 2
8742 P2004 (Address 1)

Selected Motor: Foil
2

Move Relative: Foil

Go

Current Position: Foil

Stop Motion

Status: 2

Foil Max
1

Foil Min
0

Carbon Foil Position

6000
4000
2000
0

Target Move Foil
0

Aft Plate Actuator

☐ Aft in Motion?

Selected Controller: 3
8742 P2004 (Address 1)

Selected Motor: Aft
3

Move Relative: Aft

Go

Current Position: Aft

Stop Motion

Status: 3

Aft Max
450000

Aft Min
0

Aft Plate Position

0 2000 4000 6000

Target Move Aft
0

Actuator Testing

GOAL: Check LabVIEW function

Commands tested:

- Set home (0) position
- Relative move (forward and backward)
- Get current position
- Stop movement
- Errors:
 - Trying to move out of bounds
 - Trying to move multiple actuators at a time





Critical Design Review

Data Analysis



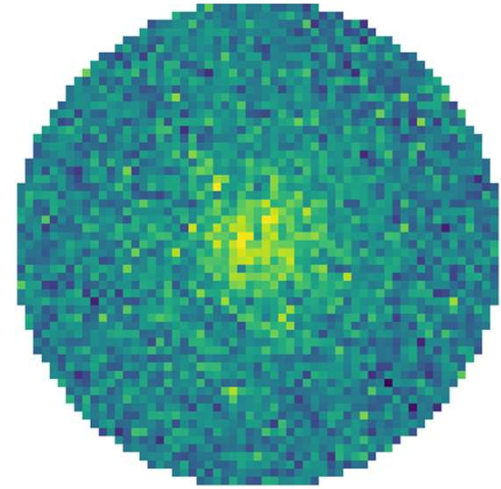
Results: Science Goal 1

1. Generate a kappa distribution model to fit the data
2. Use model to obtain FWHM
3. Scattering angle half width $\psi_{1/2} = (\text{full width half max}) / 2$
4. Convert $\psi_{1/2}$ to degrees using apparatus dimensions
5. Convert $\psi_{1/2}$ (in degrees) to thickness using the energy and atomic number:

$$t(\mu\text{g cm}^{-2}) = \frac{\bar{E}(\text{keV})\psi_{1/2}(\text{deg})}{12Z^{3/4}}$$

Results: Science Goal 1

Generate fake kappa
distribution with noise:

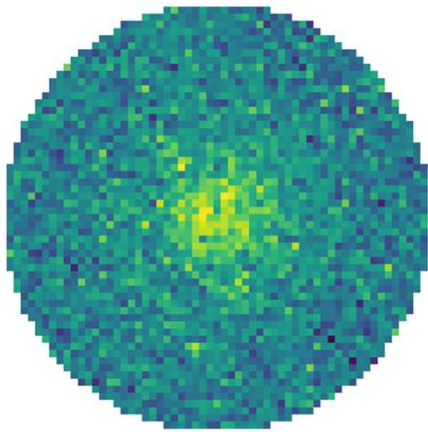


```
scipy.stats.kappa3
```

$$\kappa_3(x, y, a) = a(a + (x^2 + y^2)^{a/2})^{-(a+1)/a}$$

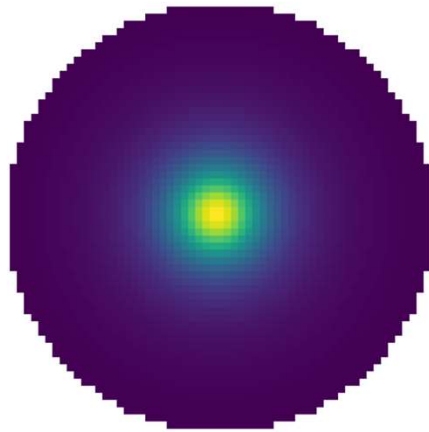
Results: Science Goal 1

Model the kappa distribution
(manually for now)



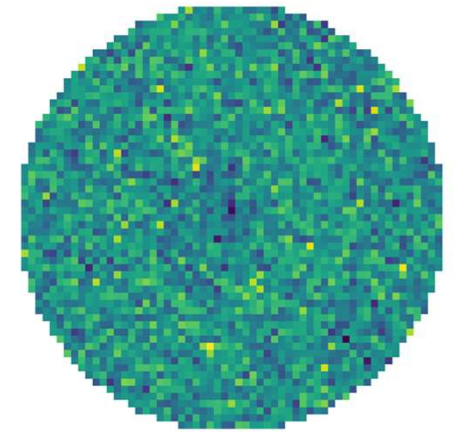
measured

—



model

=



residual

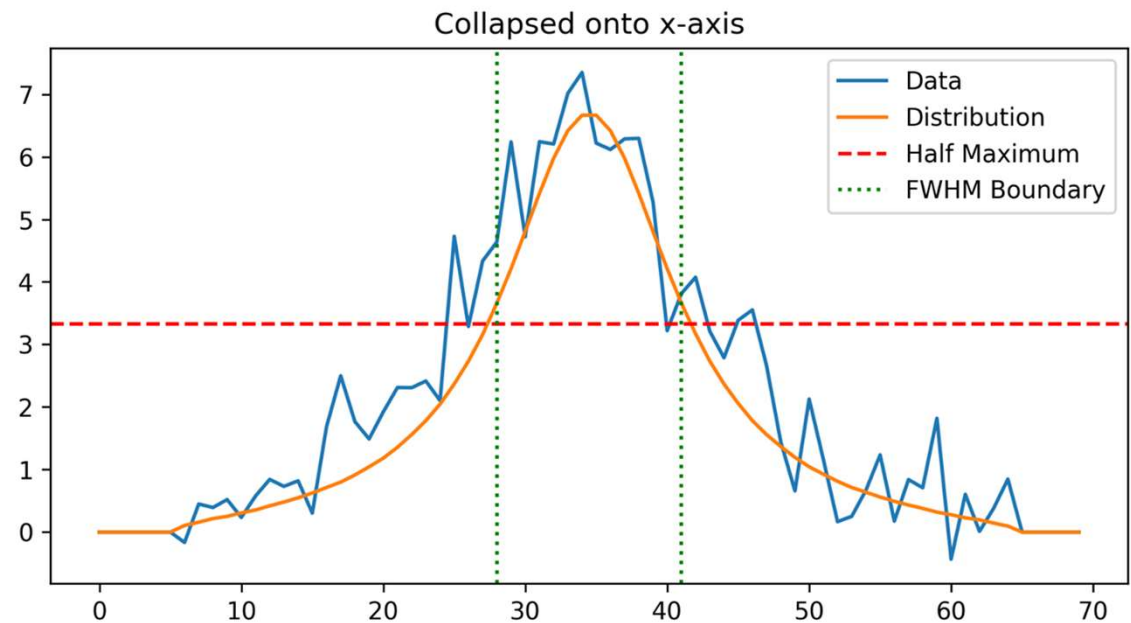
Results: Science Goal 1

Obtain $\psi_{1/2}$

In this case,

FWHM ~ 13 pixels

$\rightarrow \psi_{1/2} \sim 6.5$ pixels

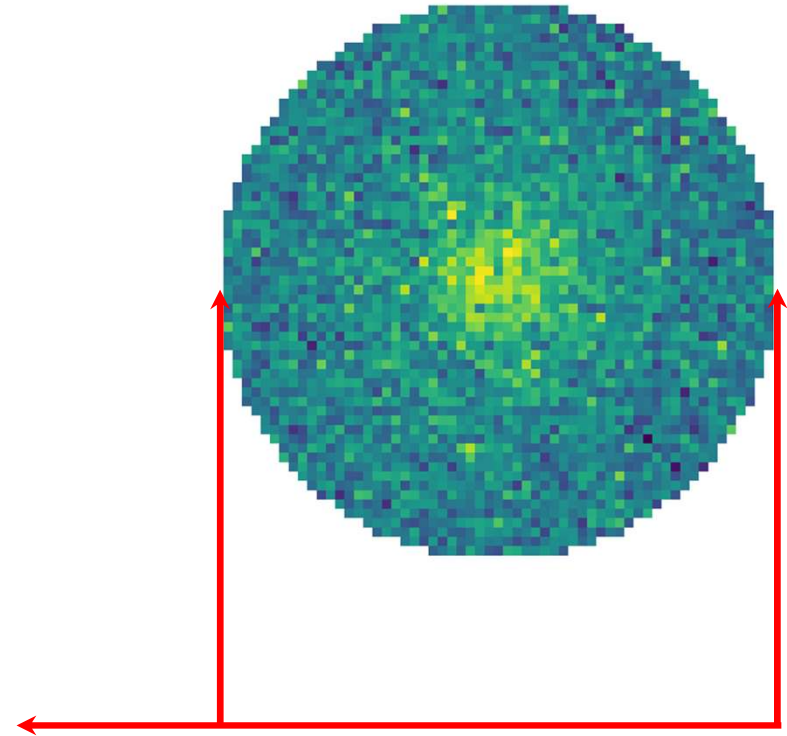


Results: Science Goal 1

Convert $\psi_{1/2}$ to physical units

60 pixels = 40mm

-> $\psi_{1/2} = 6.5 \text{ pixels} = 4.3\text{mm}$

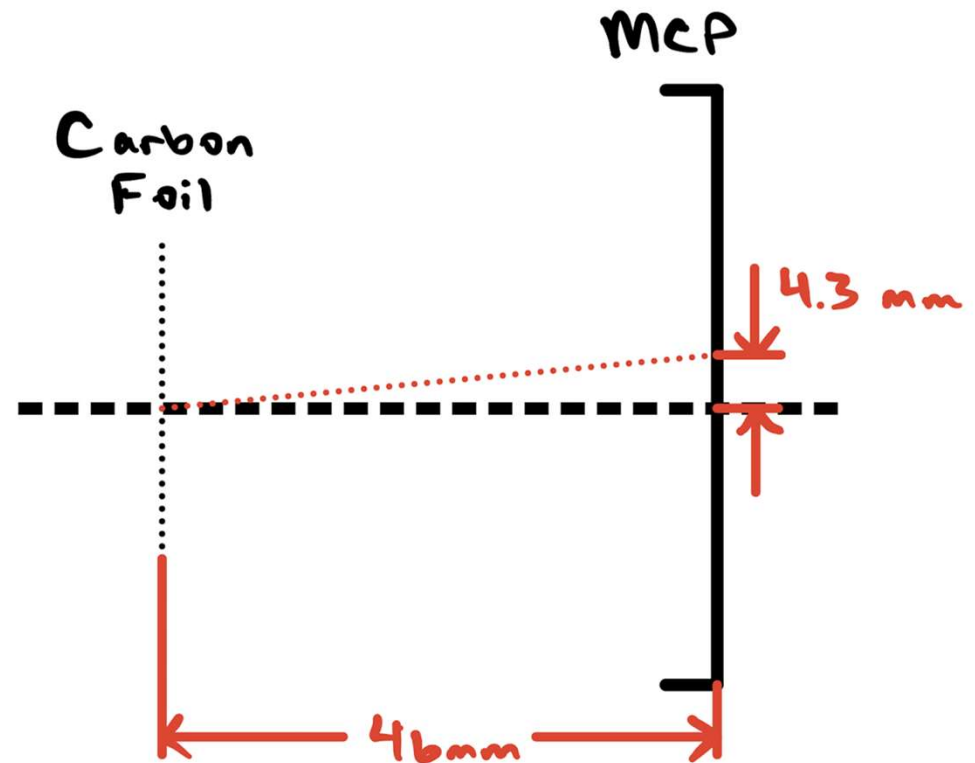


Results: Science Goal 1

Convert $\psi_{1/2}$ to degrees

$$\psi_{1/2} = \arctan(\psi_{1/2} / 46\text{mm})$$

$$= 5.34 \text{ deg}$$





Results: Science Goal 1

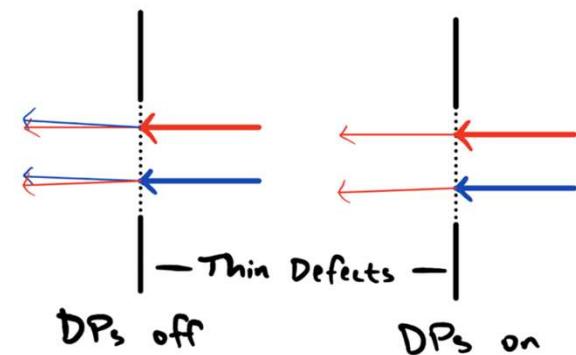
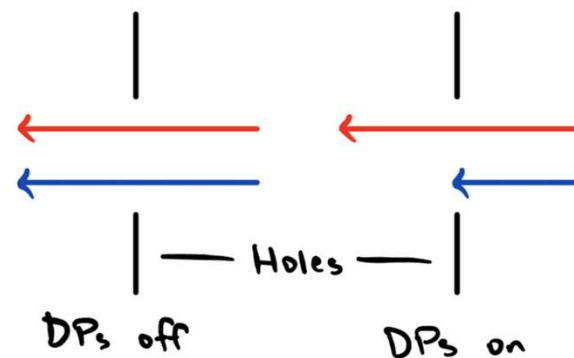
Convert $\psi_{1/2}$, beam energy (1 keV), and beam species atomic number (argon $\rightarrow$ 18) to foil thickness:

$$t(\mu\text{g cm}^{-2}) = \frac{\bar{E}(\text{keV})\psi_{1/2}(\text{deg})}{12Z^{3/4}}$$

$$\text{Foil thickness} = 0.0509 \mu\text{g cm}^{-2}$$

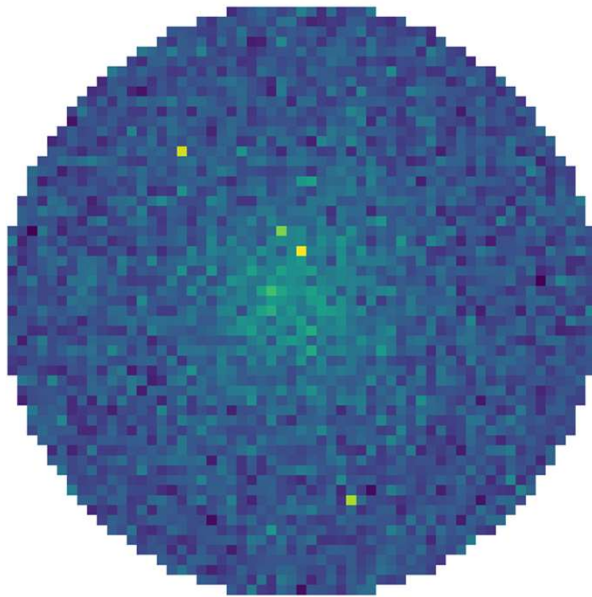
Results: Science Goal 2

1. Generate a kappa distribution model to fit the data
2. Subtract the model from the two sets of measured data: deflection plates on and off
3. Compare side-by-side the two residuals to identify peaks (above the noise) that differ between them
4. Peaks that disappear may be extremely thin defects: peaks that persist may be holes

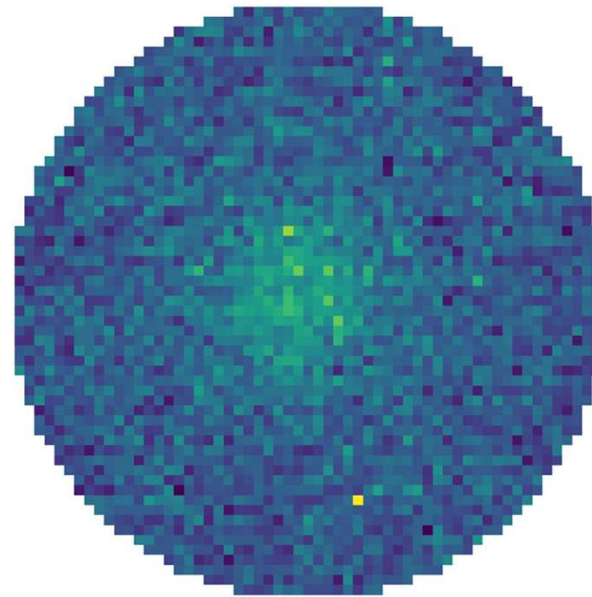


Results: Science Goal 2

Generate fake kappa distributions with noise and an anomalous pixels:



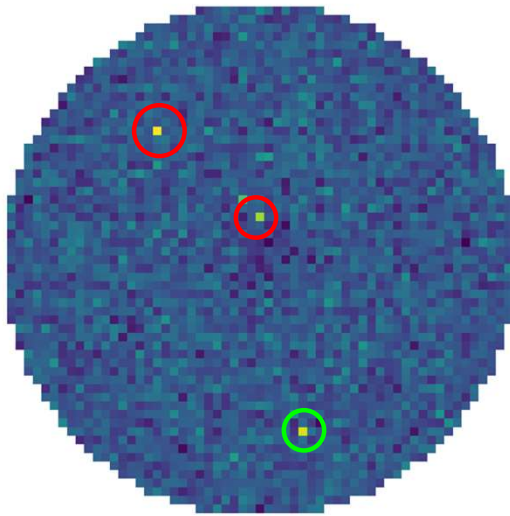
Deflection Plates Off



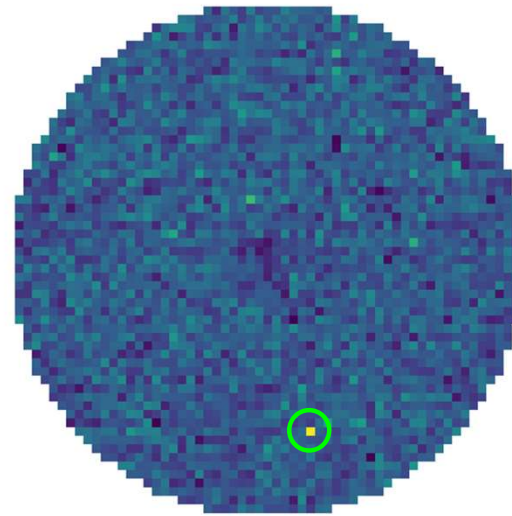
Deflection Plates On

Results: Science Goal 2

Model the kappa distribution (manually for now) and subtract



Deflection Plates
Off



Deflection Plates On

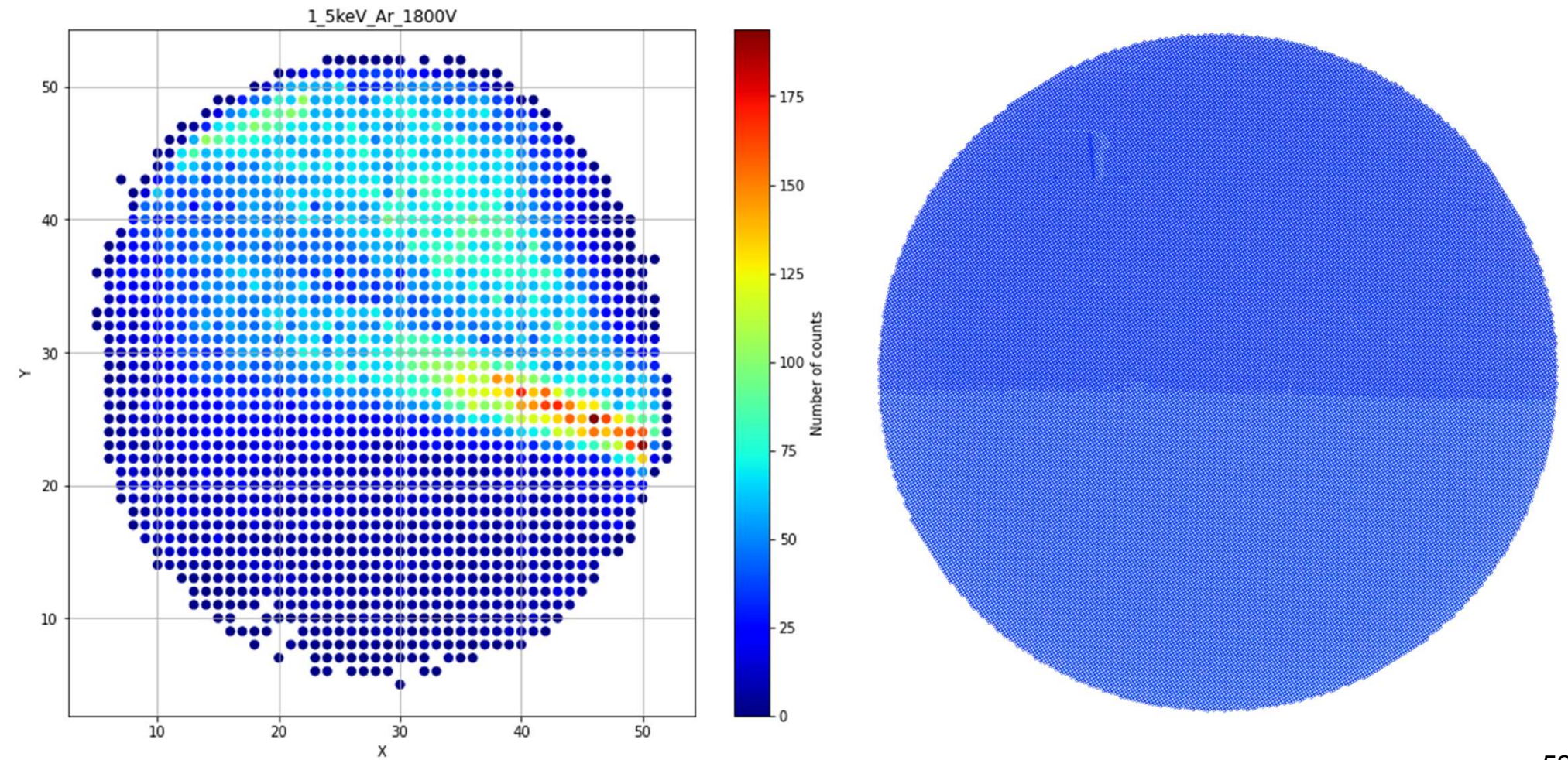
Challenge: hole/defect residuals need to be well above the noise level



Critical Design Review

Preliminary Look

Image

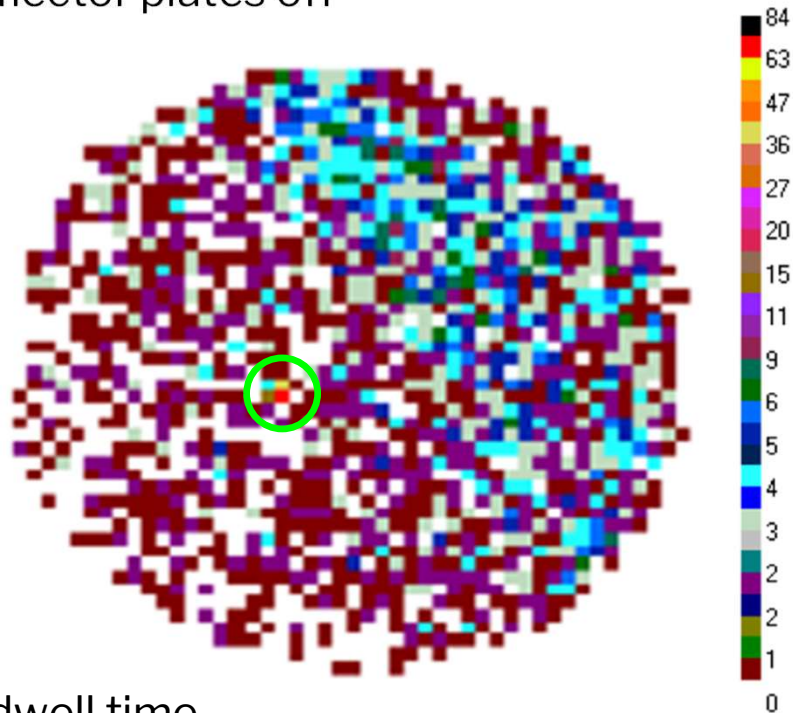


Proton

1.5 keV H⁺

Pinhole over carbon foil

Deflector plates off

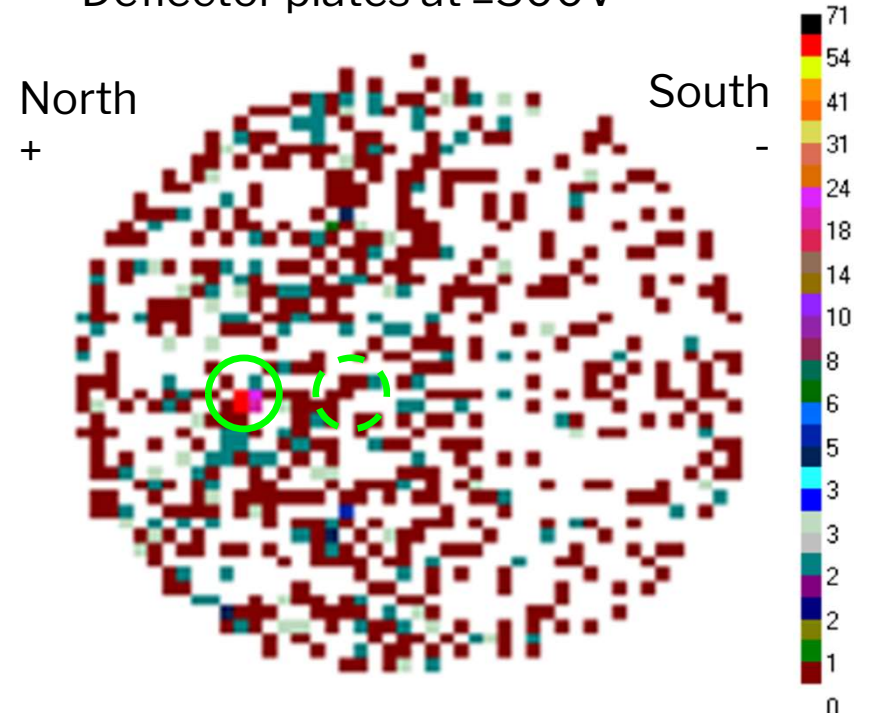


120s dwell time

1.5 keV H⁺

Pinhole over carbon foil

Deflector plates at $\pm 500V$



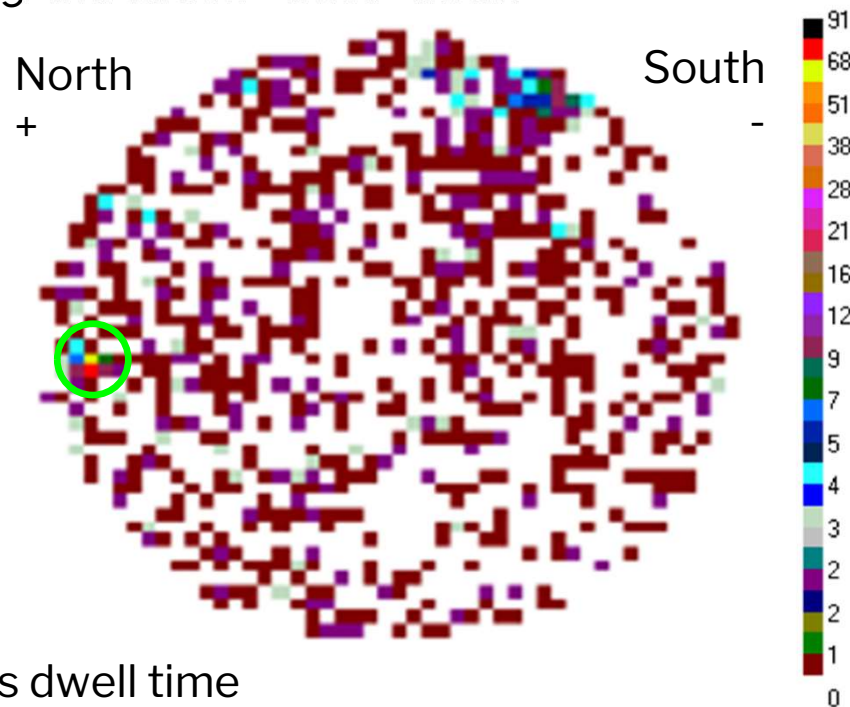
Proton

1.5 keV H⁺

Pinhole over carbon foil

Deflector plates:

1950V North 500V South

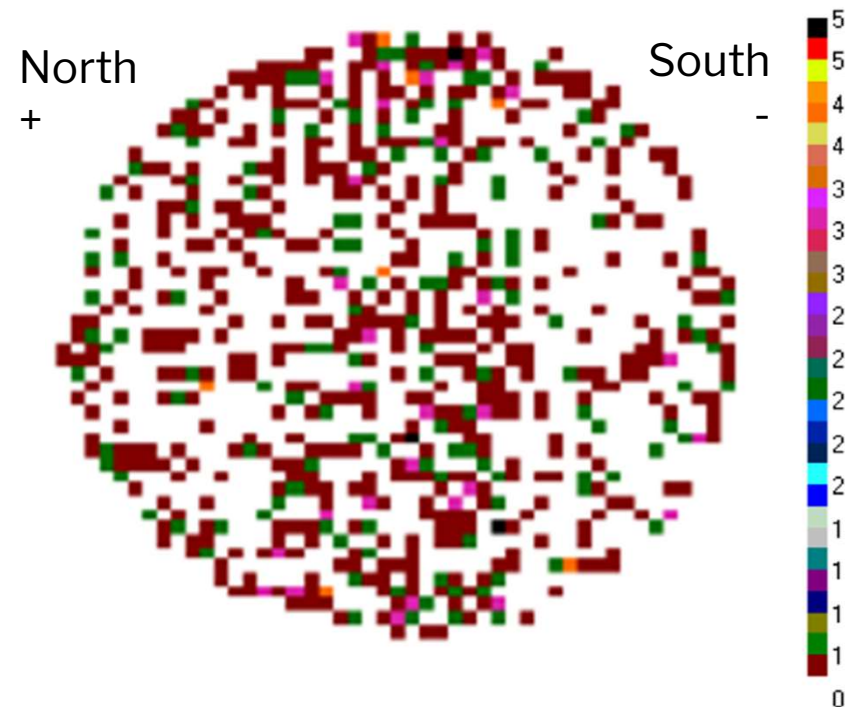


1.5 keV H⁺

Pinhole over carbon foil

Deflector plates:

500V North 1950V South

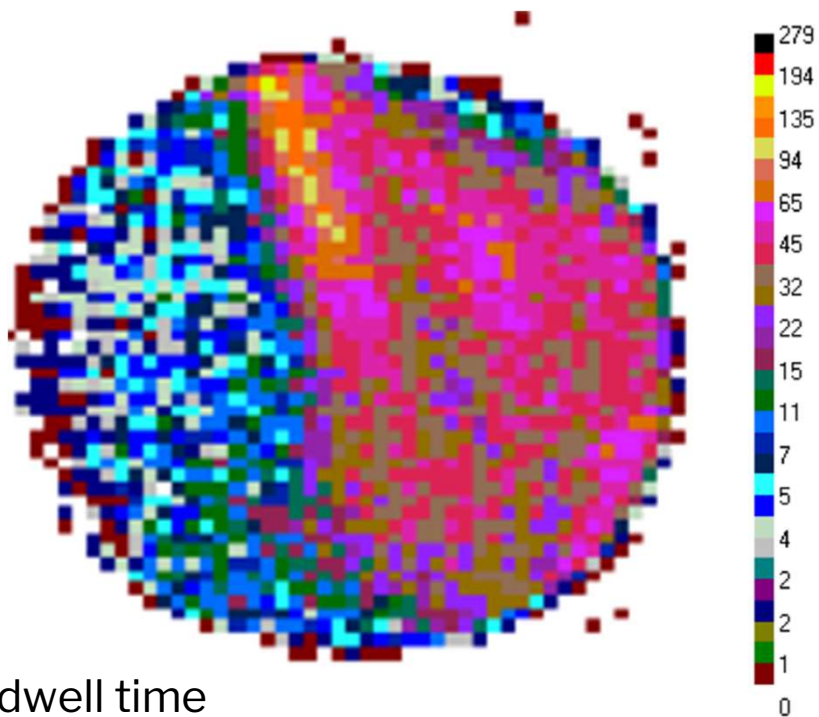


Argon

1.5 keV Ar⁺

Open foil

Deflector plates off

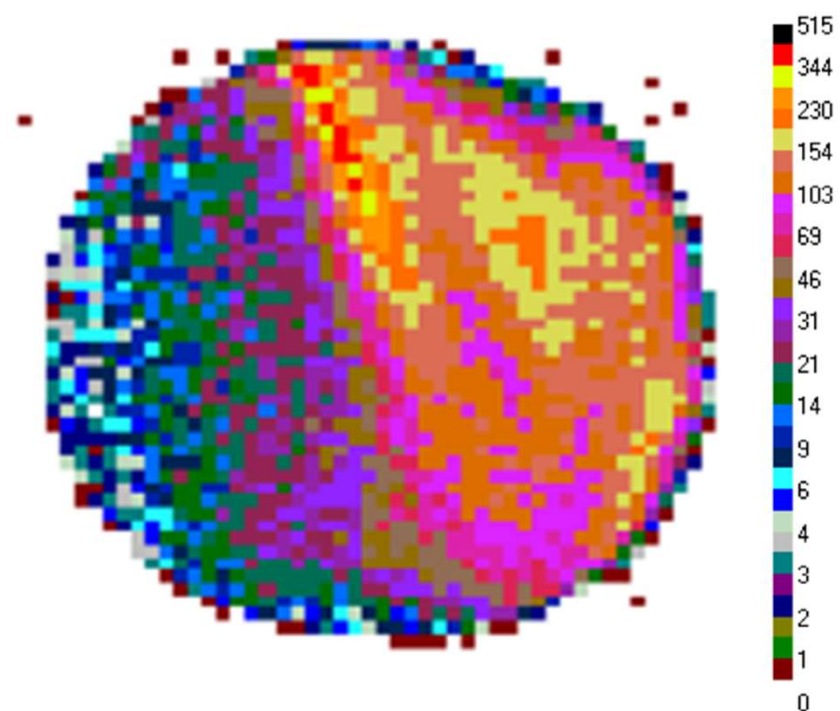


120s dwell time

1 keV Ar⁺

Pinhole over carbon foil

Deflector plates off

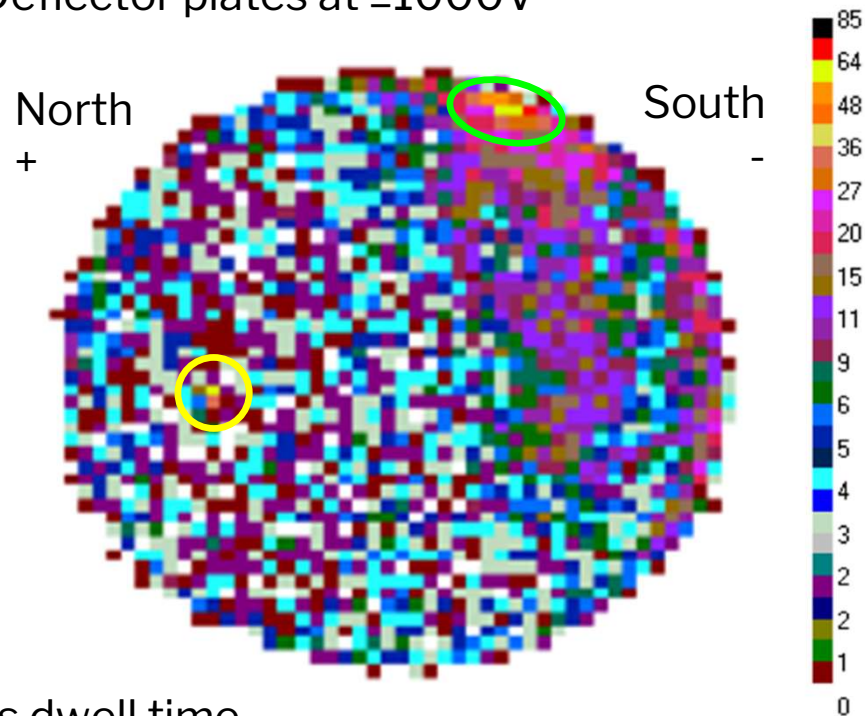


Argon

1 keV Ar⁺

Pinhole over carbon foil

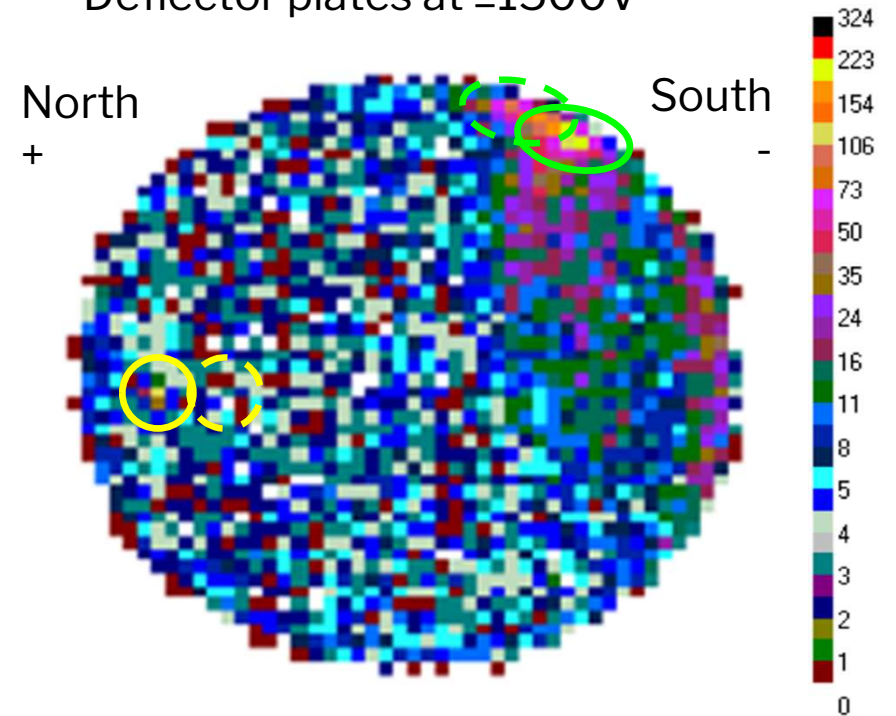
Deflector plates at $\pm 1000\text{V}$



1 keV Ar⁺

Pinhole over carbon foil

Deflector plates at $\pm 1500\text{V}$

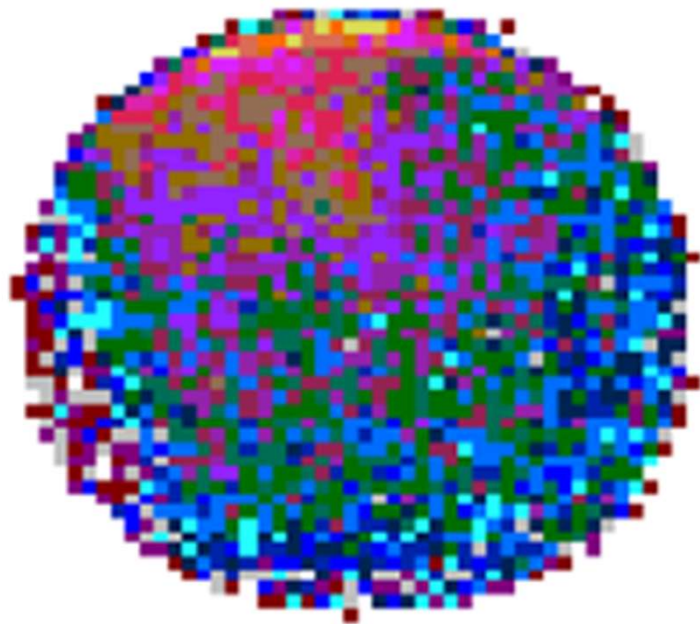


Neutral Helium

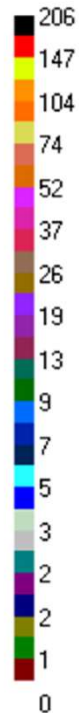
1.5 keV Charge Exchange He

Pinhole over carbon foil

Deflector plates off



60s dwell time



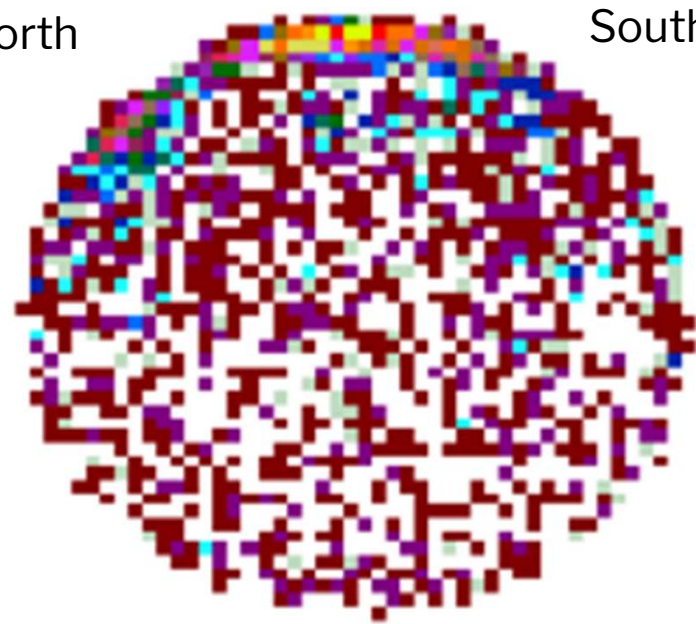
1.5 keV Charge Exchange He

Pinhole over carbon foil

Deflector plates at $\pm 1000V$

North
+

South
-



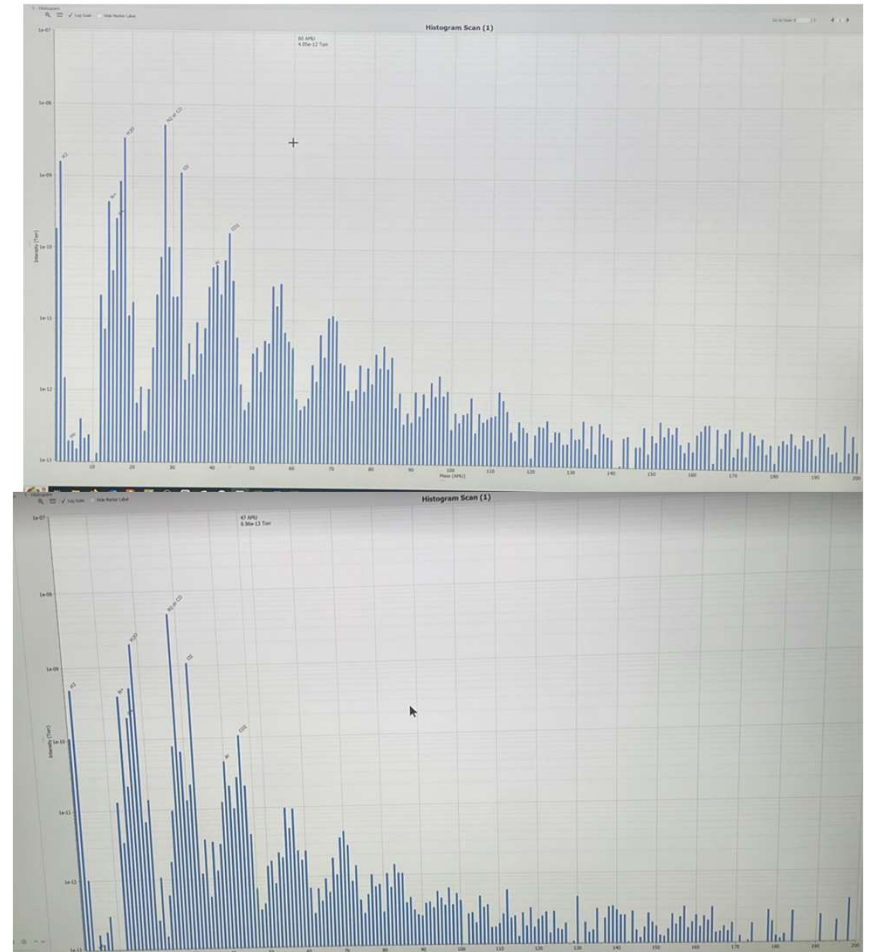


Critical Design Review

Issues

Bakeout Issue

- RGA returned dirty
- Chamber cleaned
- Thought to be dirty electrical cables
- To avoid, create a labeled clean box before commencing cleaning
- Ensures parts not sitting under cleaning fume hood before bakeout
- Reduces chance of part mix up



Slide 59

1

Describe issue in bakeout chamber + how to avoid in future classes?

this was just on the slide as a comment wanted to move it off the slide

Samuel Ruona, 5/10/2024

Actuator Calibration Issues

- Calibration of step size
 - Open loop motor has inconsistent step size
 - Mean and SD identified before bakeout, pushed back timeline
- Irregular noise during drive
 - Confirmed to be normal by Newport
 - 2000hz step velocity stated to be safe



Actuator Cleaning

- Cleaning
 - Dust found in packaging said to be clean
 - Newport recommended keeping IPA away from vacuum compatible lubricant
 - Cleaning and bakeout of actuators finished behind schedule



Actuator Drive Shafts

- Drive length inconsistent
 - Two actuators had half drive length expected
 - We decided to work with the limited length
- Temporary issue with one of the Actuators getting stuck and unable to move
 - Fix manually



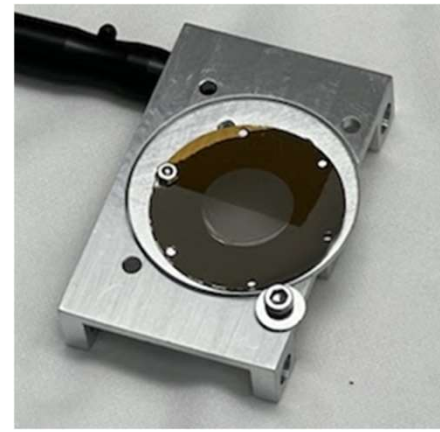
Part Fit

- Carbon foil post broke, stuck in base plate
 - Drilled out
- Actuator mounts not fitting onto actuator shaft
 - Rocked a large drill bit back and forth to widen holes
- Deflection plate screw length
 - Too short to hold washer, ring terminal, and deflection plate
 - Replaced with 8 mm screws



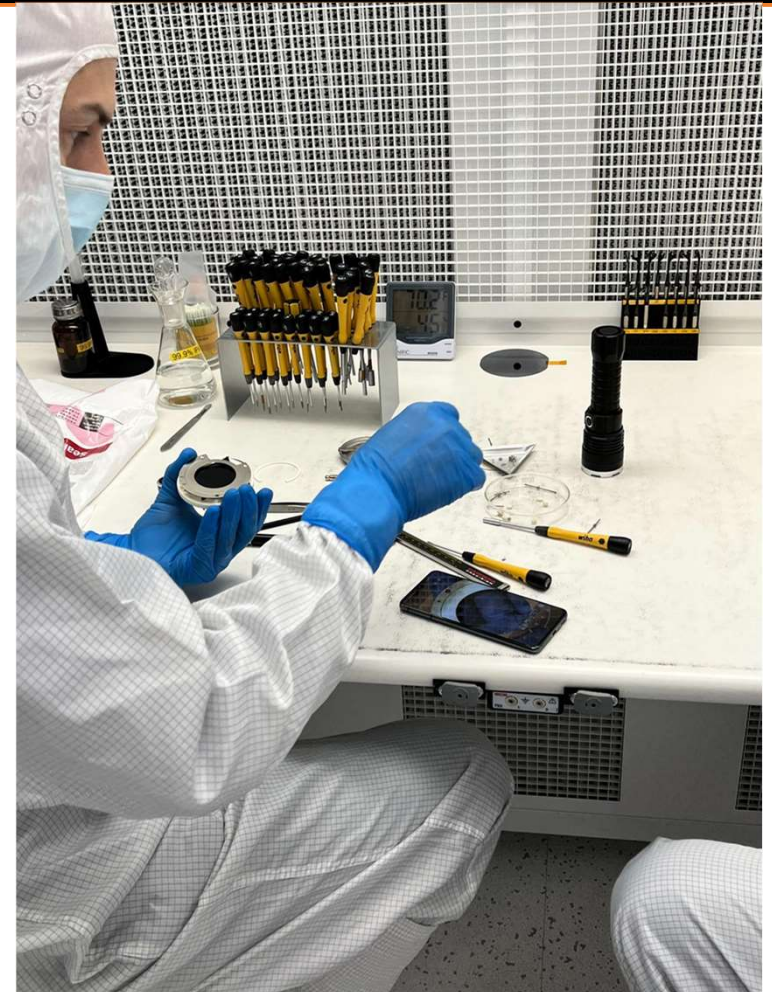
Part Fit

- Carbon foil frisbee holes didn't line up with SWAPI grid holes
 - Mounted using one hole and a washer over the edge
- MCP mounting screws too short, and fasteners would be floating
 - 2 in 1 solution - bus wire
- Screws for mounting to linear stage too narrow, and baseplate stands too thick for lab screws
 - Drilled out bigger holes and made baseplate stands thinner



MCP Calibration – Issue 1

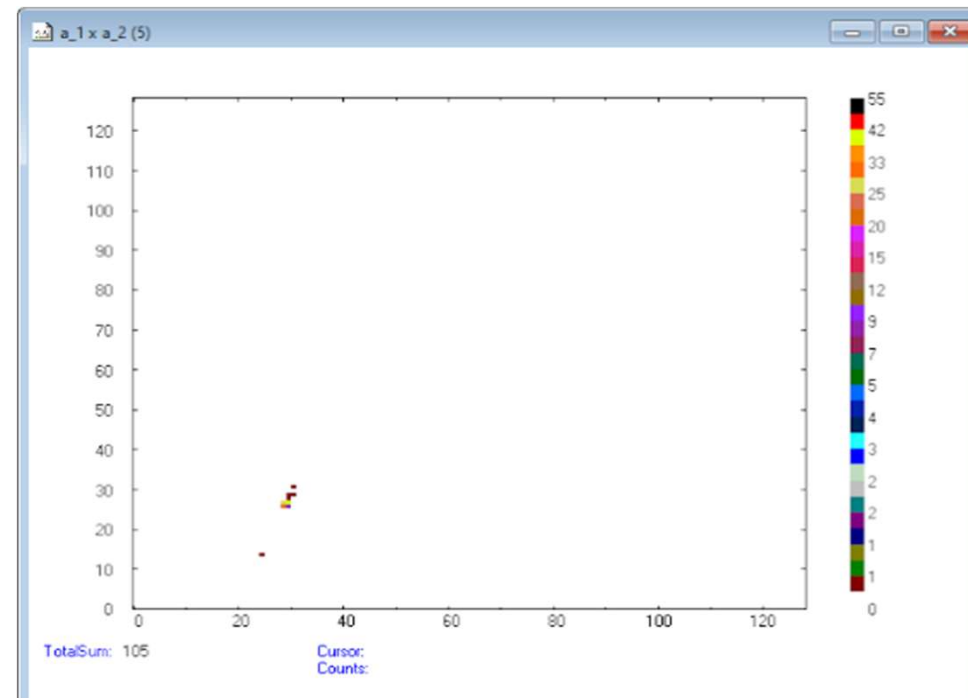
- Issue after first installation: low resistance between input and output pin, indicating that the MCP was shorted
- Fixed:
 - John widened the holes in the front ring so that the screws would not touch it



MCP Calibration – Issue 2

- After the second installation, the voltage at which we observed a significant count rate was higher than usual (2100 V)
 - Consecutive measurements while parked at 2100V were also unreliable, 132vs47 counts
- Counts were showing only on one portion of the MCP instead of having a flat field
- We observed a consistent drop in resistance of MCP, which should be pretty consistent, indicating that there was more current going through the circuit than should have

=> Conclusion was that the issue was with the spring ring, whose ‘fingers’ would relax on their own and not touch the MCP anymore





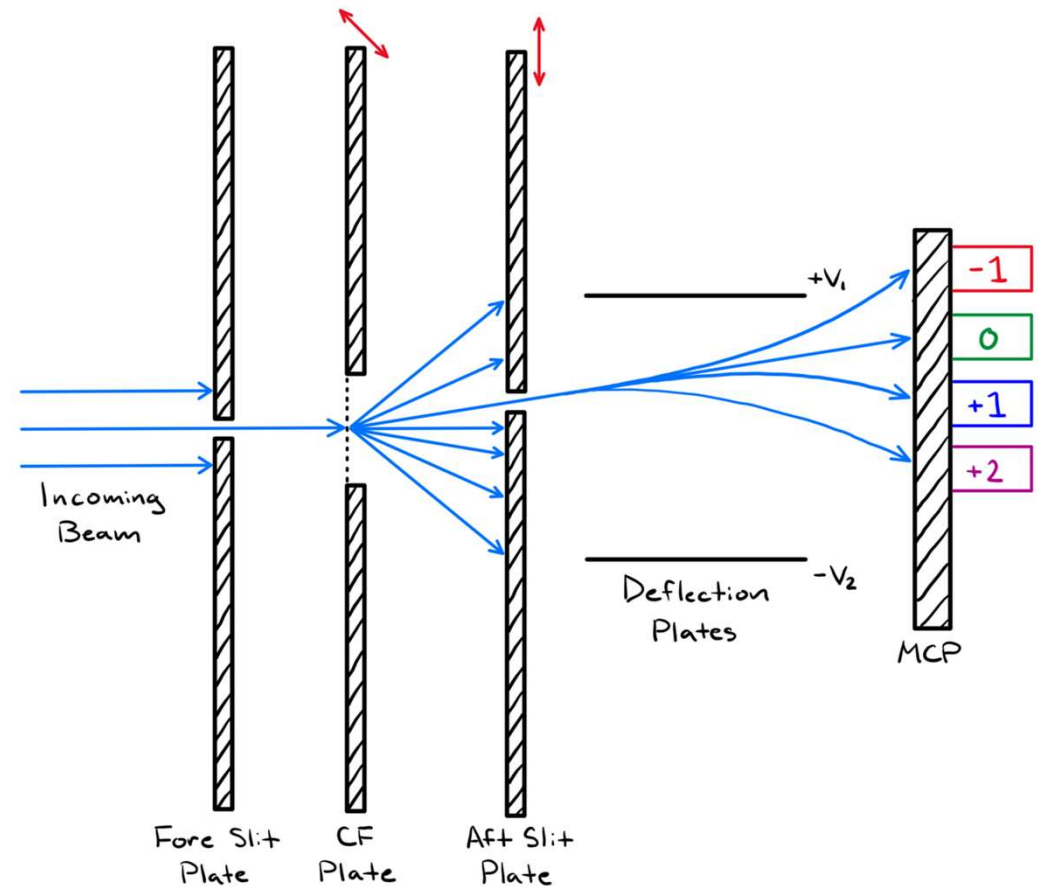
Critical Design Review

Future Steps

Further Testing

Exit charge state partitioning versus scattering angle and scattering angle half-width:

- Past studies* have examined exit charge state partitioning versus energy, species, and exit velocity
- Perhaps the neutral fraction is greatest at the center of the scattering distribution, for example?

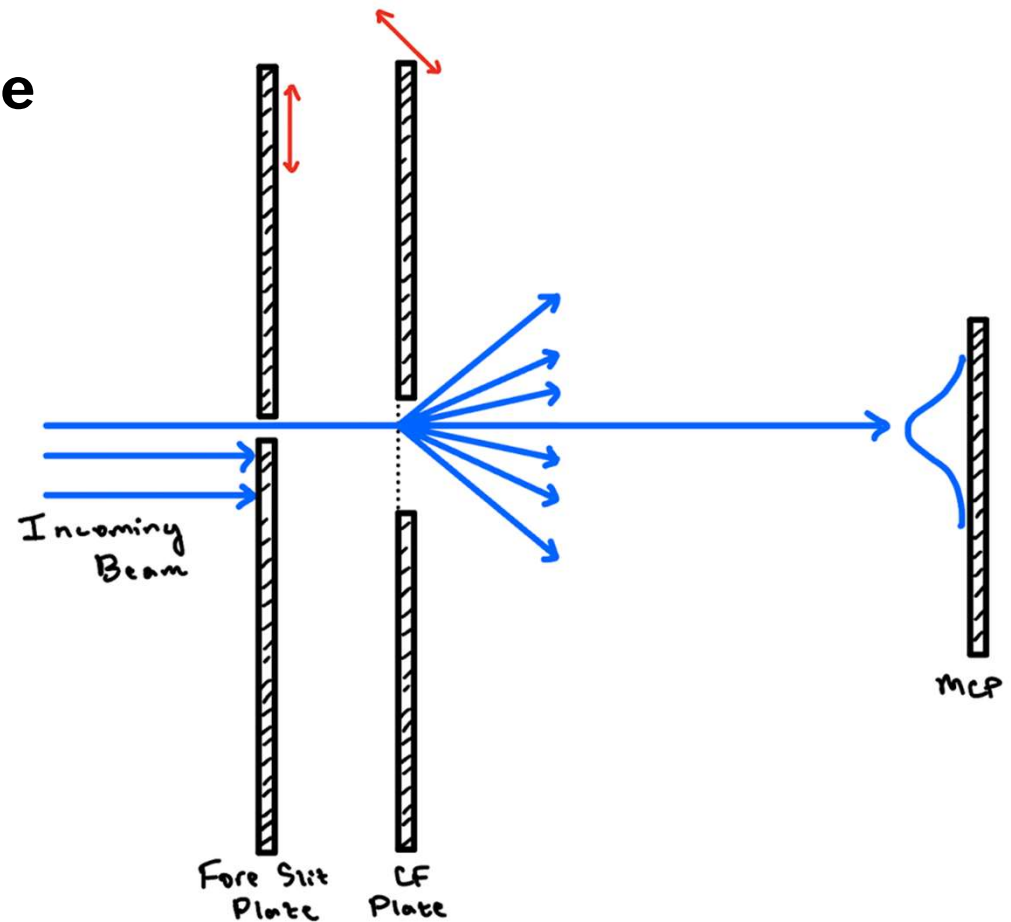


*e.g., Allegrini et al. 2014; Vira et al. 2020; Holenak et al. 2021

Further Testing

Variation of thickness across the surface of the foil:

- Past studies* have determined bulk thickness for various foils but never locally across a surface

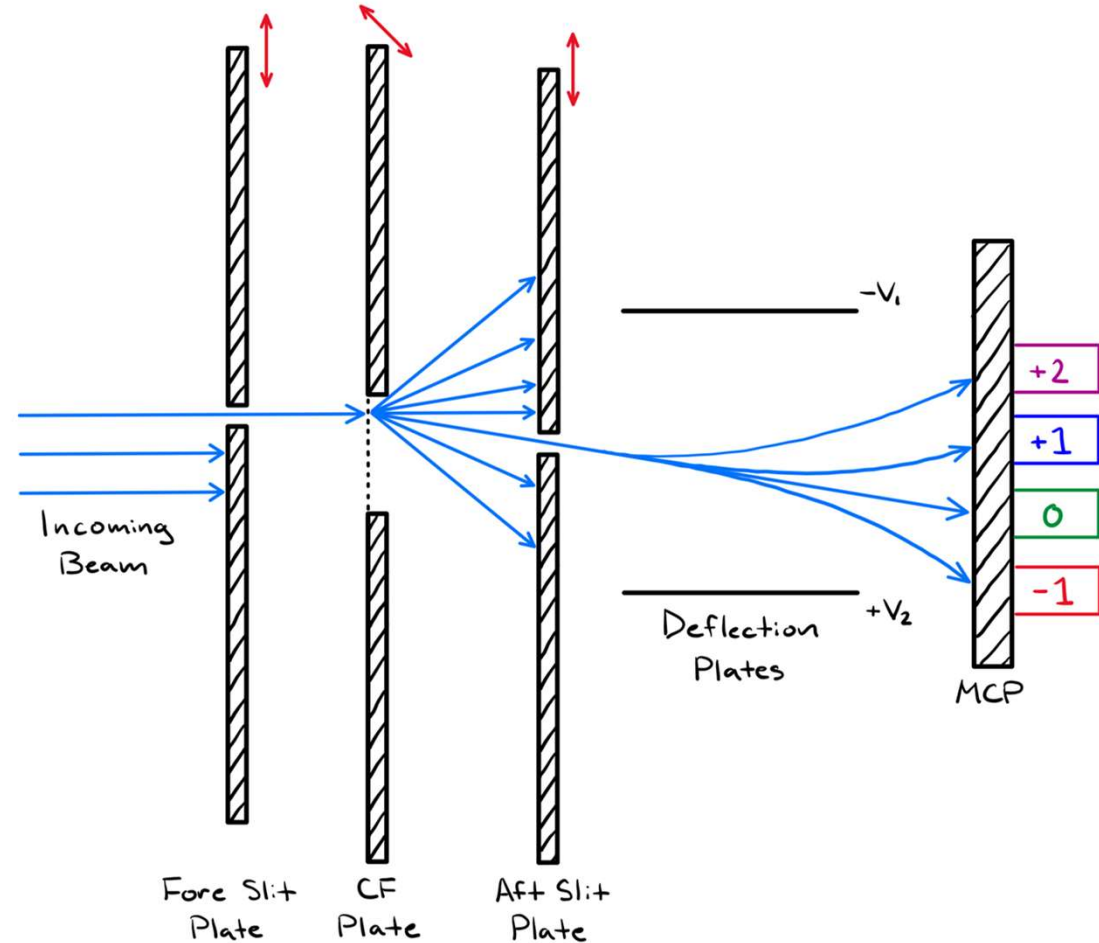


*e.g., Allegrini et al. 2016; Fhert et al. 2014; Meyer 1970

Further Testing

How does local thickness influence exit charge state partitioning for a given scattering angle?

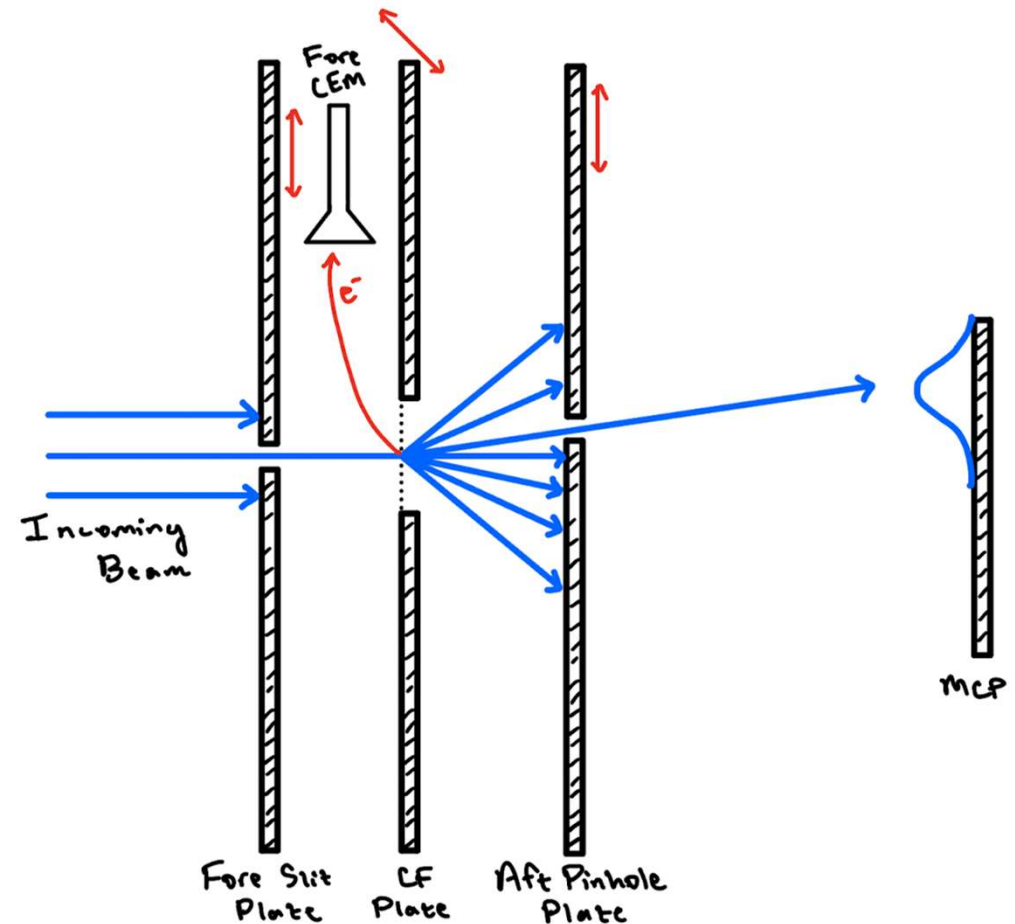
- Indirectly tells us how exit charge state partitioning varies with thickness and across the surface of a given carbon foil



Further Design Modifications

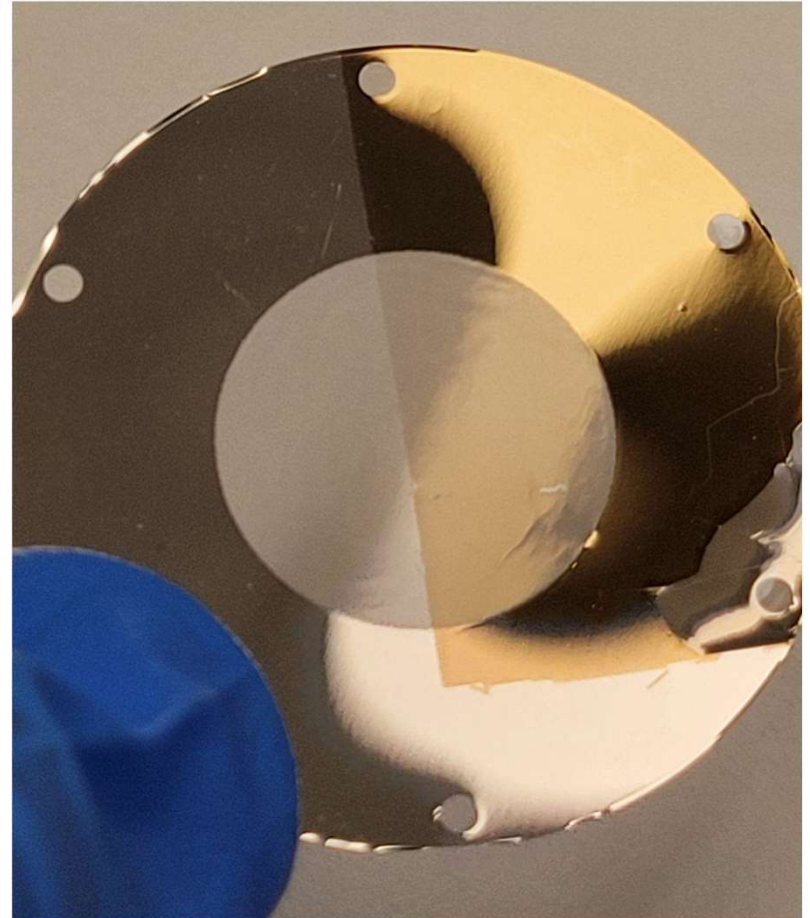
How is the rate of secondary electron emissions related to angular scattering angle?

- Past studies* have examined secondary electron yield versus energy and fluence but never scattering angle
- Perhaps ions with large scattering angles produce more secondary electrons since they pass through more carbon?



Further Possibilities

- With actuators with more range, full foils could be tested





Preliminary Design Review

Summary



Contribution to the Lab

Current Contributions:

- SIMION and CAD model of apparatus
- Assembly and Operating Procedures to Dan
- Weekly Notion progress reports
- LABVIEW code for actuator operation
- MCP plotting and data analysis code (Python)
- Utilize a designed apparatus to test Carbon Foil for pinholes and thin defects

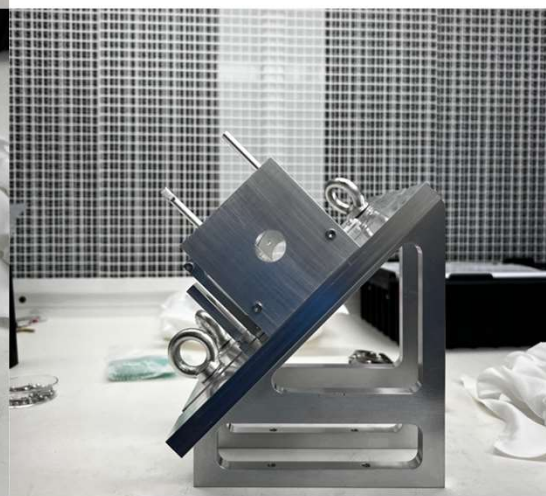
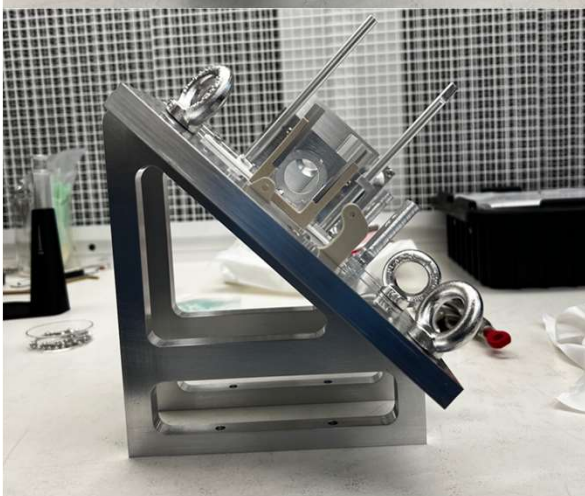
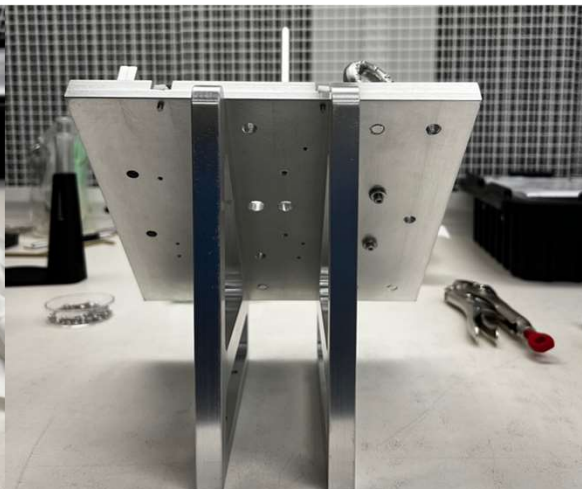


Contribution to the Lab

Future Science Contributions

Ability to...

- estimate global and local foil thickness for flight foils
- distinguish between holes and extremely thin defects
- study relationship between exit charge state partitioning and scattering angle half width
- study relationship between exit charge state partitioning and local foil thickness





Acknowledgements



Thank you everyone in Space Physics for supporting, guiding, and helping us throughout this journey! We truly appreciate you all and for the amazing lessons, knowledge, and fun gained during the two semesters!



Special Thanks!



Special Thanks to McComas, Rankin, Lengying, Mitchell, Ted, Dan, Andrew, and Brian!



Thank you!





Works Cited

- Allegrini, F., McComas, D. J., Young, D. T., Berthelier, J.-J., Covinhes, J., Illiano, J.-M., Riou, J.-F., Funsten, H. O., & Harper, R. W. (2006). Energy loss of 1–50 keV H, He, C, N, O, Ne, and Ar ions transmitted through thin carbon foils. *Review of Scientific Instruments*, 77(4), 044501. <https://doi.org/10.1063/1.2185490>
- Allegrini, F., Ebert, R. W., & Funsten, H. O. (2016). Carbon foils for space plasma instrumentation. *Journal of Geophysical Research: Space Physics*, 121(5), 3931–3950. <https://doi.org/10.1002/2016JA022570>
- Allegrini, F., Ebert, R. W., Fuselier, S. A., Nicolaou, G., Bedworth, P., Sinton, S., & Trattner, K. J. (2014). Charge state of ~1 to 50 keV ions after passing through graphene and ultrathin carbon foils. *Optical Engineering*, 53(2), 024101. <https://doi.org/10.1117/1.OE.53.2.024101>
- Ebert, R. W., Allegrini, F., Fuselier, S. A., Nicolaou, G., Bedworth, P., Sinton, S., & Trattner, K. J. (2014). Angular scattering of 1–50 keV ions through graphene and thin carbon foils: Potential applications for space plasma instrumentation. *Review of Scientific Instruments*, 85(3), 033302. <https://doi.org/10.1063/1.4866850>



Works Cited

- Funsten, H. O., Barraclough, B. L., & McComas, D. J. (1992). Pinhole detection in thin foils used in space plasma diagnostic instrumentation. *Review of Scientific Instruments*, 63(10), 4741–4743. <https://doi.org/10.1063/1.1143626>
- Holeňák, R., Lohmann, S., Sekula, F., & Primetzhofer, D. (2021). Simultaneous assessment of energy, charge state and angular distribution for medium energy ions interacting with ultra-thin self-supporting targets: A time-of-flight approach. *Vacuum*, 185, 109988. <https://doi.org/10.1016/j.vacuum.2020.109988>
- Meyer, L. (1971). Plural and multiple scattering of low-energy heavy particles in solids. *Physica Status Solidi (b)*, 44(1), 253–268. <https://doi.org/10.1002/pssb.2220440127>
- Ritzau, S. M., & Baragiola, R. A. (1998). Electron emission from carbon foils induced by keV ions. *Physical Review B*, 58(5), 2529–2538. <https://doi.org/10.1103/PhysRevB.58.2529>
- Vira, A. D., Fernandes, P. A., Funsten, H. O., Morley, S. K., Yamaguchi, H., Liu, F., & Moody, N. A. (2020). Angular scattering of protons through ultrathin graphene foils: Application for time-of-flight instrumentation. *Review of Scientific Instruments*, 91(3), 033302. <https://doi.org/10.1063/1.5134768>

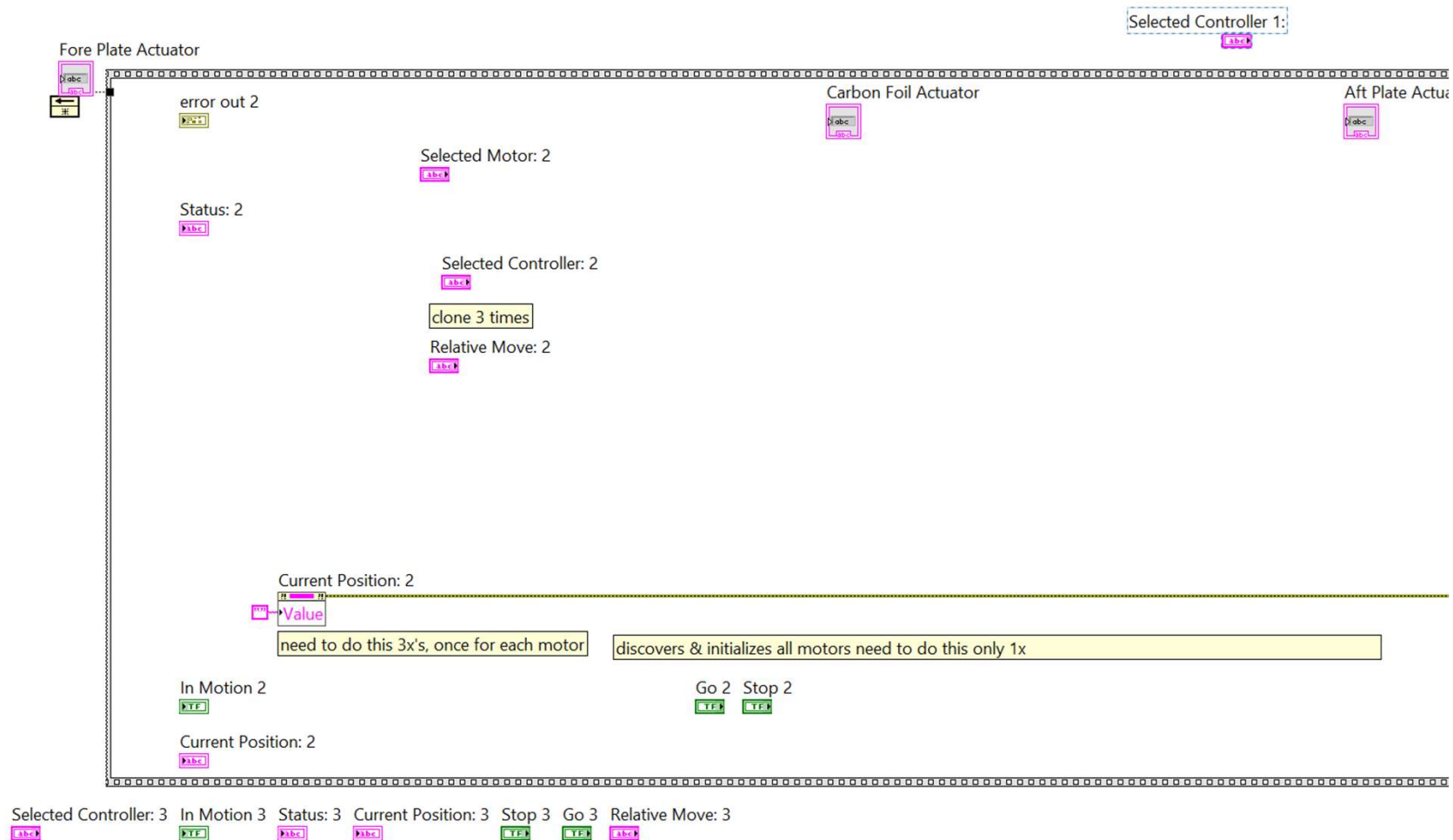


Critical Design Review

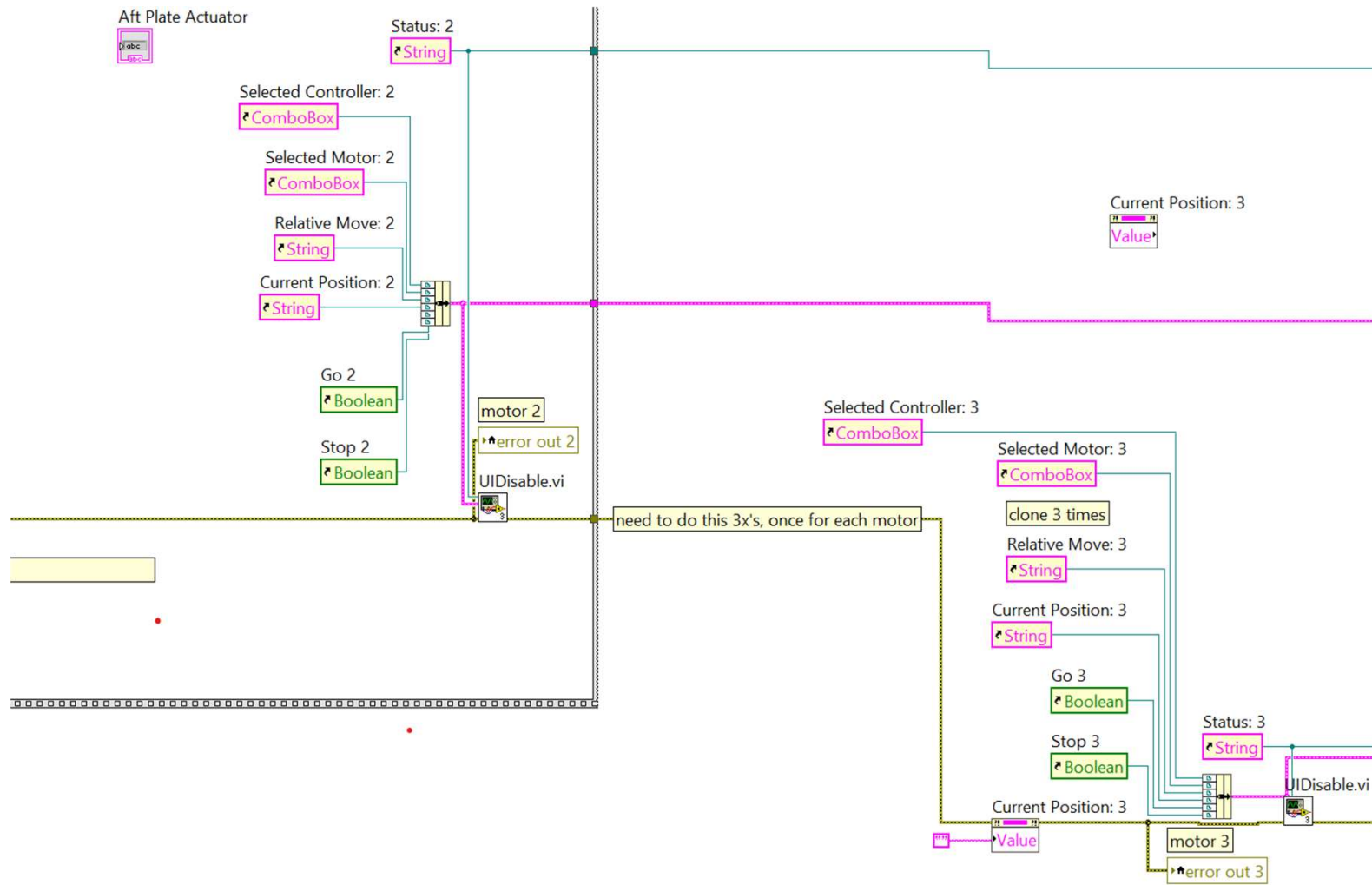
Supplemental Slides



Close Up LabVIEW Block Diagram 1



Close Up LabVIEW Block Diagram 2



Close Up LabVIEW Block Diagram 3

